Paper 3.2: RAG, PRA, and Composable Native Memory

Retrieval, Materialization, Ordering, and Positional Geometry

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Abstract

Retrieval-augmented generation (RAG) selects external evidence and serializes it into each prompt. Progressive Retrieval Attention (PRA) can instead retain a frozen selection as model-native K/V and reuse it across requests. This paper separates retrieval from that realization step, then asks when independently stored resources can be composed without recreating the packed RAG prompt. We compare lexical, dense, hybrid, reranked, and service-backed retrieval; freeze selected evidence; and test selected text, contiguous native reuse, independent records, RoPE rebinding, partial materialization, and bounded contextual repair.

Across five natural MultiHop-RAG cohorts, all 600 selector-frozen contiguous text/native comparisons preserve output, first-step logits, and answer NLL. Warm native reuse reduces TTFT by 92.3% and total latency by 81.5% relative to re-prefilling the same text, although an exact ordinary prefix-cache hit remains faster. When selections change, exact prefix reuse falls to zero while PRA reuses 28.1% of selected K/V. That saving is not free composition: independently encoded resources preserve only 8.5–10.5% of outputs, and 50% contextual repair recovers 61.5%. Partial materialization removes 25% of active K/V but lowers mean F1 from .234 to .181. BGE reranking improves packed F1 to .134, yet canonical URI-addressed records reach .080; reduced order sensitivity and non-prefix reuse do not by themselves preserve quality.

A selector-frozen causal decomposition over 150 questions localizes this gap. Removing document-to-document attention changes model behavior. Independently encoded pre-RoPE records rebound to the same packed coordinates closely track that blocked control (.108 versus .110 F1; first-step JS 8.80×10^{-4}), indicating that cross-resource contextualization, not position assignment alone, is the main conceptual difference. They are not numerically identical under the production 4-bit checkpoint: output agreement is 101/150 and value-state error grows through the decoder, although a tiny unquantized fixture matches within tolerance. Sparse boundary interaction uses far fewer cross-document edges and approaches full-packed F1, but no fixed mask dominates across F1, official score, and NLL. PRA therefore establishes exact contiguous transport and useful non-prefix reuse; changed compositions still require a qualified numerical runtime and a selective contextualization mechanism.

1 Introduction

Large language models increasingly operate over collections of documents that are too large, too dynamic, or too expensive to place directly in every prompt. Conventional RAG addresses this problem by searching an external index, selecting a small evidence set, serializing that evidence as text, and re-prefilling the model. This design has two distinct consequences. First, retrieval quality becomes a major determinant of answer quality. Second, evidence that is reused across questions is nevertheless often tokenized and prefetched through the model again unless an ordinary serving cache happens to match the new prefix.

PRA introduces a different memory boundary. Retrieved information can be stored as independently addressable resources and selected native K/V can be materialized on demand. This raises a systems opportunity—persistent reuse of evidence—but also a representation question that conventional RAG hides. A prompt such as

$$D_1\, D_2\, D_3\, Q$$

imposes a total order and a precise metric distance among documents even when D_1, D_2, D_3 are logically independent sources. If these resources are encoded separately and reused later, their original positional coordinates and contextual histories are no longer automatically equivalent to the newly serialized RAG sequence.

This paper therefore asks six linked but separable questions:

1. Which external retrieval/index strategies provide the strongest candidate evidence?
2. Given the same candidates, how do conventional RAG and PRA routing compare?
3. Given the same selected evidence, what quality–cost frontier is obtained by full text, full native, and partial PRA materialization?
4. When selected native resources are independently stored, what ordering and positional composition should they use, and how much recomputation is required to recover fresh-context behavior?
5. Does the production URI/record/resolver path alter that conclusion, and can a strong BM25–reranker selector close the remaining gap?
6. If packed RAG is denied document-to-document attention, do independently encoded pre-RoPE records rebound to the same request positions reproduce its behavior, and how much sparse interaction is useful?
7. How much residual B/C drift is attributable to arithmetic precision and shape-dependent kernels, and can request-local gist interaction add cross-document context without mutating persistent records?

The paper deliberately allows the strongest architecture to be complementary:

$$\text{mature RAG retrieval/reranking} \to \text{PRA persistent/native realization} \to \text{LLM}.$$

PRA need not beat Elasticsearch, FAISS, Qdrant, or a strong cross-encoder reranker in order to provide value.

Contributions. We make eight contributions.

1. A five-seed natural test showing exact selected-text/native transport under frozen retrieval, with ordinary prefix caching as a separate control.
2. A retrieval-index comparison spanning BM25, dense FAISS, hybrid RRF, two rerankers, Elasticsearch, and Qdrant with decomposed service latency.
3. A causal materialization ladder that contrasts score-ranked, evidence-aware, and equal-budget wrong memory while reporting active native K/V explicitly.
4. A multi-resource composition study derived from the earlier Paper 1.6 plan, separating RoPE rebinding from contextual repair and document-order sensitivity.

5. A changing-selection experiment showing non-prefix native reuse and its present fidelity cost across natural overlapping query sequences.

6. An end-to-end test of canonical PRA document records and root references with BM25, MiniLM, and BGE selection, including five-seed order and changing-query reuse measurements.

7. A selector-frozen causal decomposition that separates cross-document contextualization from pre-RoPE transport, with exact host RoPE contracts, layerwise diagnostics, and a sparse interaction ladder.

8. An explicit FP32/FP16/INT8/INT4 qualification with a shape-matched pre-RoPE control, plus parameter-free request-local gist and boundary composition mechanisms whose persistent document K/V remains immutable.

2 Relationship to Prior PRA Work

2.1 Paper 1.5: positional transport

Paper 1.5 established source-relative positional transport, deferred/pre-RoPE machinery, and rebinding algebra. It also separated positional fragmentation from contextualization fragmentation: changing the RoPE phase of a cached key can correct effective position, but it cannot reconstruct causal context that was absent when the hidden state was originally computed.

Paper 3.2 reuses that machinery and focuses on multi-resource composition rather than re-deriving the positional transport result.

2.2 The unexecuted Paper 1.6 plan

An earlier Paper 1.6 plan proposed “multi-frame positional geometry” for independently retrieved resources. It asked whether external resources need one global positional sequence and proposed source/global, globally packed, resource-adjacent, rank-distance, score-distance, near-band, and randomized policies, together with permutation, source-distance, distractor, and multi-resource experiments. That plan was not executed as a standalone paper. Paper 3.2 incorporates its central experiments in a practical RAG setting.

2.3 Paper 3.0: causal materialization

Paper 3.0 fixes selection and varies only the native K/V detail made active. It distinguishes evidence that justifies an answer to a reader from contextual states that are computationally sufficient for a particular frozen model, and establishes that encoding, search, and materialization granularities need not coincide. Its logical interval abstraction unions overlap only within one source domain, gathers across physical storage shards, and records requested, deduplicated, evidence, non-evidence, and transferred K/V separately.

Those experiments provide two constraints for the present study. First, whole-parent disclosure is a physical control, not an oracle optimum: in the controlled study, exact cores averaged 5.25 K/V tokens and reduced K/V by 92.9% relative to whole selected parents. They raised held-out answer margin by 4.107 and reached .720 accuracy, compared with a .981 margin gain and .300 accuracy for whole parents. Equal-size wrong cores instead lowered margin by .798. The matched-content intervention shows that the effect cannot be attributed to memory size alone; broader disclosure can introduce genuine attention competition rather than harmless context. Second, a small global budget is not automatically sufficient. Under oracle selection, exact disclosure reduced active K/V by 20.9% on MuSiQue and 44.4% on 2Wiki relative to whole parents, but a common 128-token allocation did not preserve all distributed MuSiQue evidence. A later Paper 3.0 audit found that this inherited pretrained replay restarted direct-query RoPE positions at zero. The comparisons remain paired within that historical transport, but Paper 3.2 does not treat their absolute quality as corrected-position evidence. These are inherited materialization findings, not Paper 3.2 RAG results. We reuse the mechanism and causal controls, then remeasure the frontier after realistic retrieval rather than oracle selection.

Paper 3.0 also found that consumer-layer placement materially changes the effect of disclosed memory, whereas contextual representation change by itself did not explain the controlled quality gap. Paper 3.2 therefore treats retrieval identity, interval disclosure, position transport, and consumer-layer use as separate factors. A RAG chunk containing the right words is not assumed to be computationally sufficient, and a compact K/V interval is not assumed to be useful merely because its evidence density is high.

2.4 Paper 3.1: retrieval addresses

Paper 3.1 studies compact persistent retrieval indices, especially generated summaries versus lexical, extractive, and native-QK alternatives. Its conclusion is kept separate. Paper 3.2 may reuse those indices as routing arms, but the present question is end-to-end RAG composition rather than summary generation.

2.5 Paper 4.5: preliminary RAG/runtime evidence

Paper 4.5 provides the initial RAG harness, candidate and selection receipts, Selected Context and Native Memory modes, and preliminary warm-reuse measurements. The present paper treats these as motivating evidence and reruns the decisive comparisons under a frozen Paper 3.2 protocol.

3 Preliminary Evidence from Paper 4.5

The preliminary measurements already suggest that two claims should be kept separate.

Strong RAG selection plus contiguous Native Memory. A strong conventional reranker selected chunks that were then consumed either as Selected Context or as PRA Native Memory. With selection receipts frozen, the Selected Context and corresponding PRA path preserved generated outputs. Reusing the same contiguous selected native block also preserved outputs.

Representative warm measurements are summarized in Table 1. These values are inherited preliminary evidence and are not yet final Paper 3.2 results.

Persistent independent chunks. A second experiment allowed selections to change and attempted reuse of independently cached chunks. The preliminary run obtained roughly 22% mean native chunk hits and reduced 101,294 selected tokens to 80,117 uniquely materialized native tokens, with cumulative runtime decreasing from approximately 151.19 s to 131.78 s. However, the official score changed from approximately 0.440 for Selected Context to 0.380 for independently cached persistent native chunks.

This distinction motivates the central composition experiment:

$$\begin{aligned}&\text{exact same contiguous block reuse}\\ &\qquad \neq \text{arbitrary recomposition of independently contextualized chunks.}\end{aligned}$$

Table 1: Preliminary Paper 4.5 warm repeated-selection results.

Model / regime	Condition	Total latency (s)	TTFT (s)
Qwen3-4B, representative	Selected Context	3.031	2.640
	Native Memory	0.638	0.166
Qwen3-4B, 4K	Selected Context	6.121	–
	Native Memory	0.799	–
Qwen3-8B	Selected Context	5.541	4.908
	Native Memory	0.990	0.276

3.1 Inherited protocol and selection boundary

The Paper 4.5 protocol used 50 fixed MultiHop-RAG questions (seed 11) from a 609-document corpus. BM25 produced immutable candidate receipts at $N \in \{5, 10, 20, 50\}$; a 256-token chunker with 32-token overlap then supplied either global BM25 packing or parameter-free PRA rank fusion. A broad probe measured answer-string availability in selected context, not generated-answer quality. At the 2K budget and $N = 20$, PRA increased availability from .56 to .68, supporting-document coverage from .430 to .493, and span coverage from .268 to .318, while false selected-document fraction rose from .769 to .777. At $N = 50$, availability increased from .64 to .76, but support coverage changed from .515 to .505, span coverage from .338 to .300, and false-document fraction from .776 to .795. The result is a budget-sensitive selection signal, not evidence of better grounded generation.

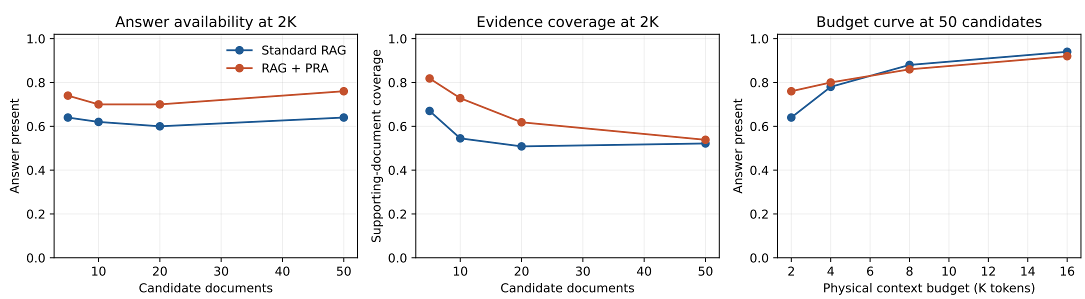

Figure 1: Inherited Paper 4.5 selection probe. The vertical measure is whether selected context contains an accepted answer string; it is not model accuracy. The source artifact remains preliminary for Paper 3.2.

3.2 Inherited powered decomposition

The powered 4B run used the same 50 questions, $N = 20$, a 2K budget, pinned Qwen3-4B 4-bit weights, and a fixed cross-encoder. Standard BM25 obtained .500 official score and .082 token F1. The stronger conventional selector obtained .440/.081, and generic PRA obtained .440/.071. The generic selector nevertheless increased supporting-document coverage from .400 to .468 and gold-chunk recall from .0448 to .0484. Thus evidence exposure and generated-answer quality moved in different directions, closing the inherited selection gate.

The matched realization result was positive and narrower. Strong conventional and strong-PRA Selected Context produced identical outputs; within generic and strong PRA, Selected Context and Native Memory also matched all generated outputs. Generic cold native execution cost 3.121 s versus 3.024 s for selected text, a 3.2% regression. Reusing the exact native selection reduced mean TTFT from 2.640 to 0.166 s and total latency from 3.031 to 0.638 s, while carrying approximately 299 MB of active all-layer K/V. The 4K boundary retained exact realization, paid a 3.3% cold regression, and reduced warm total latency from 6.121 to 0.799 s at roughly 602 MB active K/V.

The prespecified 8B replication preserved the distinction. Standard BM25, the stronger conventional selector, and generic PRA scored .520/.093, .480/.096, and .440/.081, respectively. Both native realization pairs remained output-exact. Cold native execution was 1.8% slower (5.636 versus 5.535 s), whereas retained native reuse reduced total latency from 5.541 to 0.990 s and TTFT from 4.908 to 0.276 s, using approximately 285 MB of active detail.

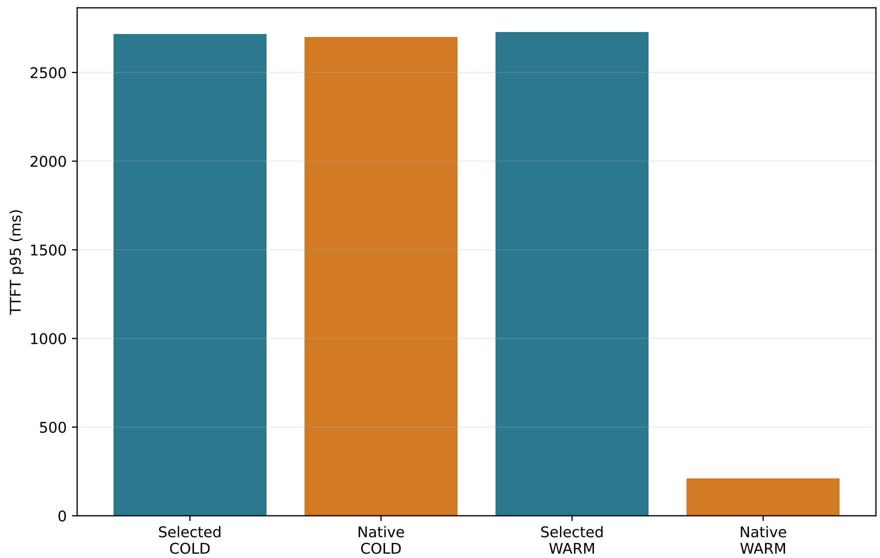

Figure 2: Inherited selector-frozen Qwen3-4B TTFT. Cold and retained-selection regimes are disjoint; the warm result is a repeated-selection benefit, not an unconditional RAG speedup or a comparison against an ordinary prefix-cache hit.

Finally, the independent-chunk run changed both geometry and reuse unit. It reused 22.0% of chunk lookups, reduced approximately 101,294 selected tokens to 80,117 uniquely materialized tokens, and lowered cumulative wall time from 151.19 to 131.78 s. Official score changed from .440 to .380 while token F1 changed from .071 to .086. Independently encoded chunks retained source-local post-RoPE positions instead of the fresh selected text's packed global positions. Paper 3.2 therefore treats this row as the motivating composition failure, not as transport parity or cache corruption. All numbers in this section are inherited from commit `f60b6847eb65f8a5be9c758a34a2ebdc09244e6e` and remain preliminary until reproduced here.

4 Paper 3.2 Mechanism Gate

Before loading a model, we ran the complete receipt and geometry pipeline on 15 controlled multi-document questions under five corpus seeds. Three first-stage backends produced 225 immutable candidate receipts; external-order and PRA-refined selections produced 450 selection receipts; six composed profiles and seven applicable position policies produced 4,050 composition receipts. Every receipt round-tripped with a stable digest, and all position policies preserved resource identity, source hash, token count, and internal token offsets.

At document Recall@10, BM25, exact hashed-dense retrieval, and RRF hybrid obtained .833, .900, and .913. Their external-order selected-document recalls were .587, .667, and .733. PRA second-stage ranking reached .733 for all three candidate sets. Figure 3 shows the decomposition. The synthetic fixture is deliberately lexical and small, so these numbers validate information flow and candidate ceilings; they do not rank production retrievers.

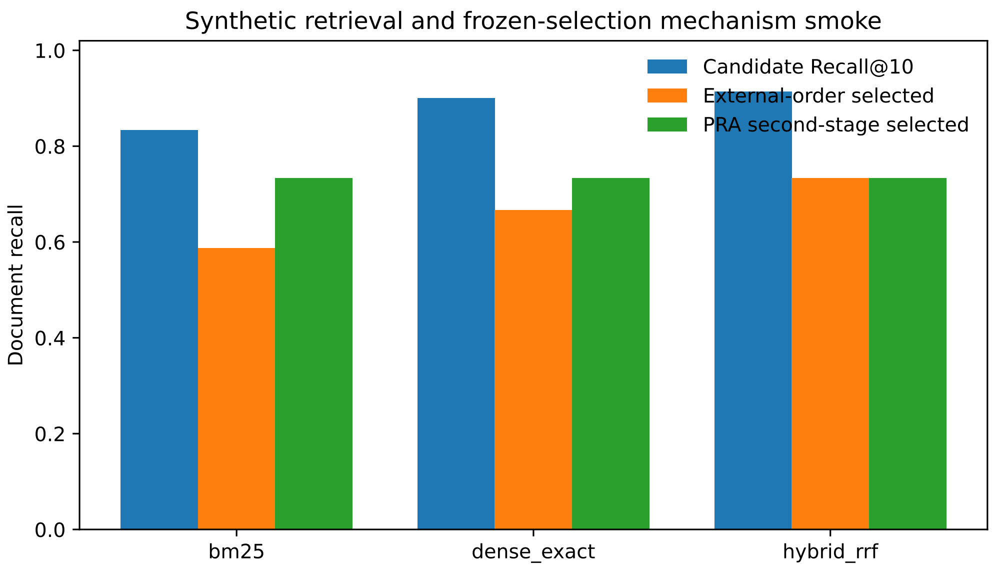

Figure 3: Five-seed synthetic retrieval and frozen-selection mechanism gate. No language model is run, and answer-quality metrics are intentionally absent.

The coordinate check also distinguishes overlap from identity collision. Globally packed, rank-distance, random-distance, and non-overlapping near-band policies assign disjoint effective positions. Source-local and resource-adjacent policies intentionally overlap approximately .807 of numeric coordinates on this fixture; score-distance overlaps .455. In every case the resource URI and chunk identity remain part of the native address, so equal position numbers from different documents are not deduplicated. This gate qualifies the experiment machinery only. Model-backed permutation, rebinding, and repair results are reported in the natural composition study below.

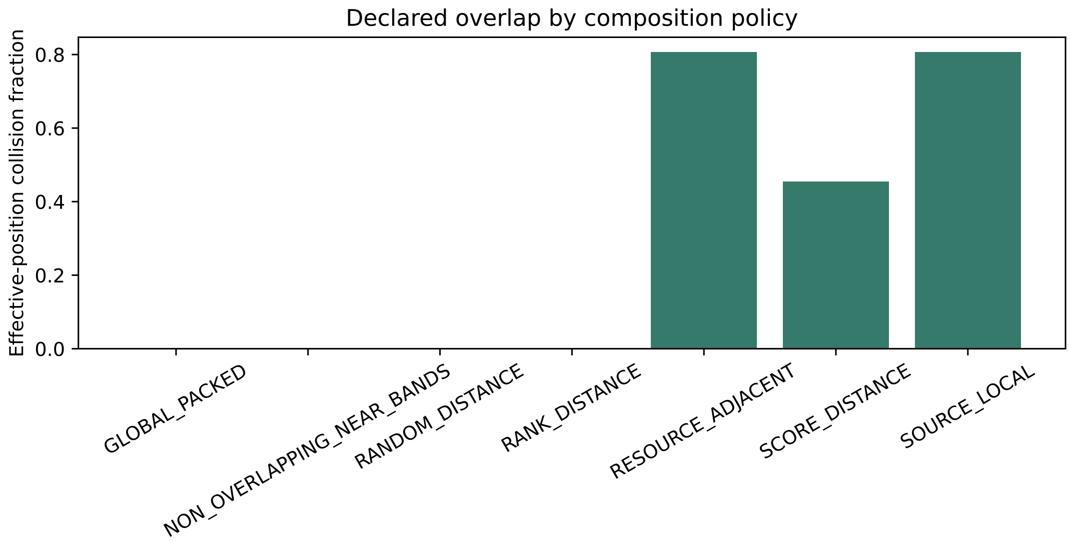

Figure 4: Effective-coordinate overlap by declared composition policy. Numeric position overlap is intentional for source-local and adjacent-resource frames; logical resource identities remain distinct.

4.1 First model-backed transport gate

We next ran the selector-frozen realization path with Qwen3-1.7B-4bit on an Apple M4 Pro. The model revision was pinned to `3b1b1768`; no PRA adaptor was available. Each of five corpus seeds used the same 15 controlled questions, while distractor construction varied by seed. We therefore use seed, rather than question execution, as the replication unit. The experiment produced 450 measured rows and 150 matched Selected Context/Native Memory pairs across cold and warm regimes.

All 150 pairs shared the same candidate receipt, selection receipt, selected intervals, and token-F1 value, and all 150 generated strings matched exactly. Table 2 reports means across the five seed summaries. Native realization replaced approximately 122 visible evidence tokens with the same number of selected native K/V states; the direct visible prompt fell from 153.6 to 32.0 tokens. Cold native execution remained 3.3% slower in TTFT and 4.1% slower end to end. Once the exact selection was retained, native ingestion fell from 60.5 to 0.04 ms, TTFT from 102.2 to 37.0 ms, and total latency from 177.8 to 116.5 ms.

Table 2: Five-seed controlled Qwen3-1.7B transport gate. Values are means of 15-question seed summaries; parentheses give between-seed standard deviation.

Regime	Realization	F1	visible	TTFT (ms)	total (ms)
Cold	Selected text	.094 (.010)	153.6	100.6 (1.7)	176.2 (2.7)
Cold	Native K/V	.094 (.010)	32.0	103.9 (2.5)	183.5 (2.7)
Warm	Selected text	.094 (.010)	153.6	102.2 (1.8)	177.8 (2.5)
Warm	Native K/V	.094 (.010)	32.0	37.0 (0.4)	116.5 (0.5)

This result closes only the narrow transport gate. Standard BM25 achieved mean token F1 .109 and supporting-document coverage .933, while the generic PRA selector obtained .094 and .873. The fixture is lexical, small, and repeated across seeds; exact match was zero for every arm because

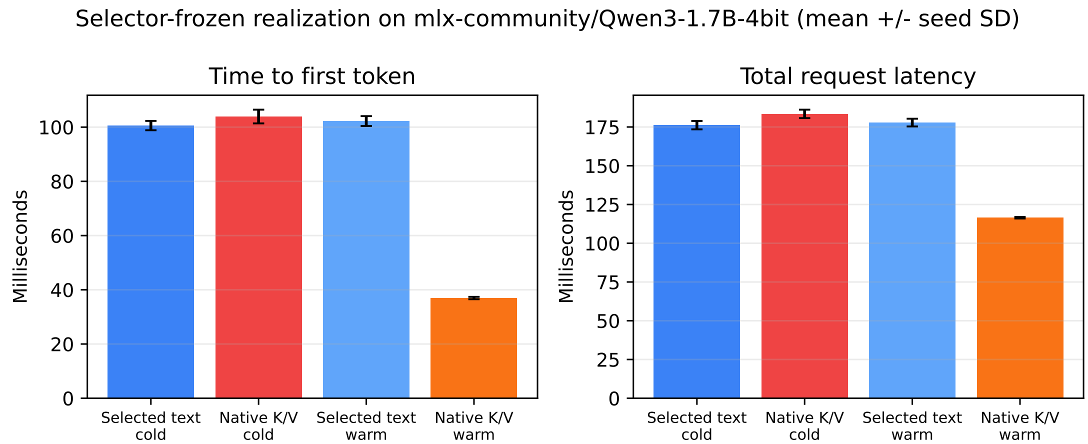

Figure 5: Matched realization cost on the controlled fixture. Error bars are between-seed standard deviations. Native reuse is beneficial only in the warm regime here; the cold path pays native encoding and is slightly slower.

the model returned sentences rather than the requested short value. Accordingly, the selection gate remains closed on this fixture.

4.1.1 Natural MultiHop-RAG replication

We then ran the same pinned model on 50 sampled MultiHop-RAG questions at the inherited $N = 20$, 2K-token operating point, with a cross-encoder pinned to revision `233902d2`. This is one fixed cohort (seed 11), not five independent natural cohorts. All 200 cold/warm Selected Context/Native Memory pairs—100 generic and 100 cross-encoder-selected—again shared candidate and selection receipts, intervals, scores, and exact generated output. Table 3 separates selection from realization.

Table 3: Qwen3-1.7B on 50 MultiHop-RAG questions. "Vis./native" reports direct visible prompt tokens and selected native K/V tokens.

Condition	Regime	official	F1	support	vis./native	TTFT	total
BM25 text	Cold	.480	.119	.400	2108/0	1013	1105
Cross-encoder text	Cold	.500	.137	.455	2110/0	1020	1111
PRA generic selected	Cold	.440	.122	.468	2105/0	1000	1092
PRA strong selected	Cold	.500	.137	.455	2110/0	1003	1095
PRA strong native	Cold	.500	.137	.455	79/2031	1005	1129
PRA strong native	Warm	.500	.137	.455	79/2031	73	196

Generic PRA raised answer-string availability from .56 to .64 and supporting-document coverage from .400 to .468; gold-chunk recall moved from .0448 to .0484. The official score nevertheless fell from .48 to .44, while token F1 changed from .119 to .122. The cross-encoder was stronger: answer availability .70, official score .50, and token F1 .137. Its ordinary RAG, PRA Selected Context, and PRA Native Memory outputs agree exactly. The result therefore supports conventional retrieval followed by PRA realization, but does not establish PRA as a better first- or second-stage selector.

Cold strong-selector native execution was 3.1% slower end to end than its matched Selected Context arm, while TTFT changed by only 0.2%. Retaining the exact selection reduced TTFT from 999 to 73 ms (92.7%) and total latency from 1091 to 196 ms (82.0%), at 222.1 MiB of active all-layer K/V.

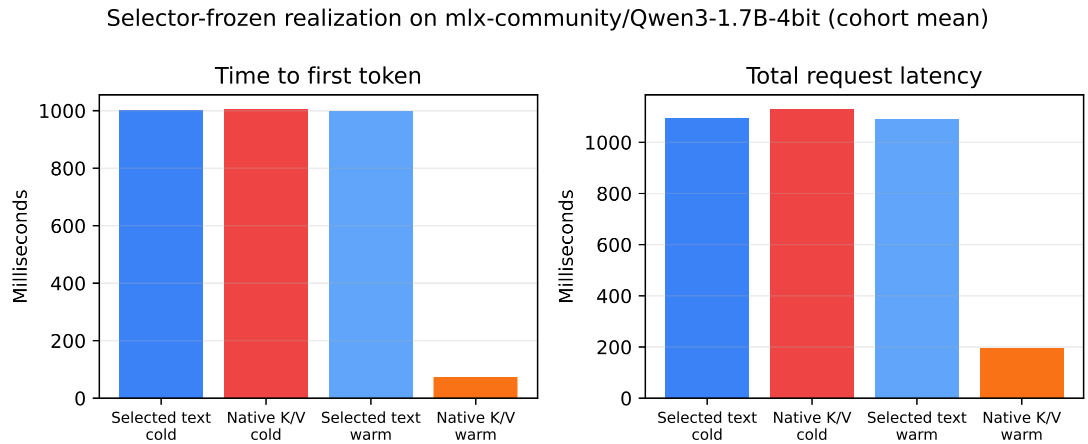

Figure 6: Single-cohort MultiHop-RAG realization cost with selector output frozen. No between-seed error bars are available for this first natural run.

The native and repeated-selection economic gates pass at the prespecified 50-example threshold. The selection gate fails because generic PRA does not match the strongest conventional selector. The public-card gate also remains closed because no exact model/precision adaptor is qualified.

We then replicated transport on five independently sampled 30-question natural cohorts (seeds 11, 23, 37, 71, and 101). Across generic and strong selectors and cold and warm execution, all 600 matched text/native pairs preserved selection receipts, generated output, first-step-logit hashes, and token F1; the maximum gold-answer NLL difference was zero. Table 4 reports the strong-selector arm. Relative to repeated selected-text prefill, retained native K/V reduced warm TTFT by 92.3% and total latency by 81.5%. The exact prefix cache was still 6.9% faster in TTFT and 18.9% faster end to end than native reuse. This is the intended boundary: PRA does not improve an unchanged serialized prefix; its target is reuse when evidence recurs outside that prefix.

Table 4: Five-seed natural transport confirmation (30 questions per seed, strong reranker). Quality and support are identical within every matched text/native pair.

Realization/regime	F1	Support	Visible/native	TTFT (ms)	Total (ms)
Selected text, cold	.116	.542	2109/0	1007	1103
Native K/V, cold	.116	.542	80/2029	1016	1143
Selected text, warm re-prefill	.116	.542	2109/0	1006	1100
Native K/V, warm	.116	.542	80/2029	77	203
Selected text, exact-prefix hit	.116	.542	2109/0	72	165

The five-seed answer-log-probability and independently composed-resource experiments below extend this transport result; a reduced Llama-family replication is also reported.

5 Experimental Questions

5.1 Full context, RAG, or PRA?

When all candidate documents fit the native context window, we compare:

1. **Full-Doc-In-Context**: all candidate documents are freshly encoded;
2. **conventional RAG**: retrieved chunks are freshly packed as text;
3. **PRA full materialization**: PRA routes but exposes all tokens of selected resources;
4. **PRA partial materialization**: PRA exposes a bounded selected subset;
5. **PRA Native Memory**: selected evidence is represented by reusable native K/V.

Full context is a reference rather than a guaranteed upper bound: retrieval may improve quality by removing distractors.

5.2 Does PRA need to replace the RAG retriever?

No such assumption is made. We test:

$$
\begin{aligned}
&\text{external retrieval} \to \text{ordinary RAG},\\
&\text{external retrieval} \to \text{PRA second-stage selection},\\
&\text{strong reranker} \to \text{PRA Selected Context},\\
&\text{strong reranker} \to \text{PRA Native Memory}.
\end{aligned}
$$

This explicitly separates selection quality from realization and reuse.

5.3 RAG+PRA as a composed system

We represent RAG+PRA as a sequence of independently auditable stages:

$$
\begin{aligned}
&\text{query and corpus revision} \to \text{external retrieval/index} \to \text{candidate receipt}\\
&\to \text{optional PRA second-stage routing} \to \text{selection receipt} \to \text{materialization and composition}\\
&\to \text{Selected Context or Native Memory} \to \text{persistent reuse} \to \text{generation}.
\end{aligned}
$$

External retrieval may use BM25, dense/FAISS search, hybrid fusion, a strong reranker, or a retrieval service. Its immutable candidate receipt records source identities, spans, scores, ranks, and index revision. The subsequent selection receipt fixes the exact evidence consumed by matched realization conditions. In particular, a native-memory condition cannot rerun retrieval and still be described as a transport comparison.

5.4 Canonical PRA record contract

The record experiment reuses the Paper 4.5 runtime contract rather than an experiment-specific document type. Each source is a versioned `ContextRecord` of type `generic_document`; its payload retains the source URI, document metadata, sections, and immutable chunk descriptors. Each chunk is a child `rag_chunk` record with source-relative character offsets and a stable URI of the form `pra://multihoprag/document/`d`#chunk=`i. Records carry the same selection policy, named views, source fingerprint, and serialization boundary used by the runtime SDK. Ingestion receipts also use the semantic fields of the Paper 4.5 `PRAStorageEntry`: logical key, record type, retention class, resource version, reconstructability, source URI, and source hash. Engine-specific model, topology, data-type, layout, position-policy, and consumer-profile fields are bound only when native K/V is encoded.

Request-specific ranks and reranker scores never mutate persistent records. They remain in candidate and selection receipts. An explicit request resolves leaf document references; a routed request resolves one request-scoped `rag_chunk_set` root and then applies the same frozen selection. Reference tokens are removed before model tokenization. Both paths persist the logical prompt, resolved record and chunk URIs, visible model prompt, selection-receipt identity, and hashes of materialized text. This separation allows the measured record behavior to be merged into Paper 4.5 without a record-schema translation layer.

Table 5 defines the composed profiles used throughout the study. These profiles let us test PRA in three distinct roles: replacement selector, second-stage selector, and realization/reuse layer after a conventional selector.

Table 5: Canonical RAG+PRA profiles. Conditions sharing a selection receipt hold evidence fixed and change only its realization or composition.

Profile	Selection	Realization and purpose
`RAG_ONLY_TEXT`	External RAG	Freshly packed visible text; strong conventional baseline.
`RAG_PLUS_PRA_SELECTED`	External RAG, optionally PRA-refined	Fresh Selected Context; isolates second-stage routing without native transport.
`RAG_PLUS_PRA` `NATIVE_CONTIGUOUS`	Frozen external selection	One reusable packed native block; isolates reuse under ordinary packed geometry.
`RAG_PLUS_PRA` `NATIVE_INDEPENDENT`	Frozen external selection	Independently cached source-position K/V; exposes composition error.
`RAG_PLUS_PRA_NATIVE_REBOUND`	Frozen external selection	Independent K/V rebound to request-specific composition positions; isolates positional repair.
`RAG_PLUS_PRA_REPAIR`	Frozen external selection	Rebound K/V plus bounded contextual repair; estimates the recomputation needed for fresh-context parity.

The first comparison is always `RAG_ONLY_TEXT` against the realization-only native profiles with identical evidence. We then evaluate external retrieval followed by PRA refinement, and only afterward PRA as a replacement selector. Thus a retrieval improvement, a materialization saving, and a positional-recomposition result remain separate empirical claims.

6 Datasets

The study uses a staged dataset ladder.

Mechanism and inherited QA. Controlled synthetic multi-document facts provide exact causal checks. Natural PRA cohorts include HotpotQA, QASPER, 2WikiMultiHopQA, and MuSiQue.

Flagship RAG. MultiHop-RAG is the mandatory first flagship dataset because preliminary Paper 4.5 infrastructure already exists. Once the protocol is stable we add KILT Natural Questions and KILT HotpotQA.

For each dataset we freeze corpus and query revisions, official metrics, chunking, candidate document construction, evidence labels where available, and the subset for which Full-Doc-In-Context fits the model's native window.

7 Model Scaling

The full experimental grid is intentionally run only on small models.

Stage M0. Qwen3-0.6B is the primary mechanism model. Llama 3.2-1B provides an early cross-family check where practical.

Stage M1. After correctness gates, a 1–2B class model is used for reduced replication.

Stage M2. Qwen3-4B is mandatory because it connects directly to Paper 4.5 preliminary RAG evidence. Qwen3-8B is the first scaling replication.

Stage M3. Only the strongest conditions are repeated on Qwen3-14B and, if compute permits, a larger Qwen and at least one non-Qwen family.

This staged design tests whether an apparent mechanism survives scale without multiplying every retrieval, position, and materialization factor at every model size.

8 Retrieval and Index Strategies

8.1 Local/custom indices

We begin with transparent in-process baselines.

BM25. A pinned BM25 implementation provides the lexical reference.

Dense semantic retrieval. The requested "FAS" path is interpreted as FAISS. We use one frozen embedding model and compare exact dense search on small corpora with a pinned FAISS ANN configuration on larger corpora.

Hybrid. Lexical and dense candidates are fused first using reciprocal-rank fusion (RRF). A predeclared weighted fusion may be added as a secondary arm.

Strong conventional reranker. A frozen cross-encoder or equivalently strong conventional reranker provides both a competitive RAG baseline and a selector-frozen input to PRA realization.

8.2 Real retrieval services

We deployed pinned Elasticsearch and Qdrant services and evaluated lexical, dense, and service-level RRF retrieval against the same corpus and query receipts. Additional database vendors were deliberately deferred: the purpose is not a vendor benchmark, and Elasticsearch plus one vector database is enough to test whether the retrieval conclusions survive a service boundary. The user-specified "Derby" name was not silently interpreted as Apache Derby or a vector service without a pinned product and API.

The purpose is not to publish a standalone vector-database benchmark. Instead we ask whether RAG/PRA conclusions survive realistic retrieval infrastructure. Service latency is decomposed into search, network, reranking, and LLM costs.

9 PRA Routing and Materialization

9.1 Routing families

Within an identical candidate corpus, candidate PRA routing modes include:

- lexical routing;
- native semantic/gist routing;
- native-QK or the strongest inherited low-rank QK selector where available;
- lexical–semantic hybrid routing;
- the strongest inherited generic PRA profile;
- optional Paper 3.1 summary indexing as a secondary arm.

9.2 Materialization families

We distinguish:

Full. Materialize all tokens from each PRA-selected object.

Fixed partial. Materialize a fixed chunk/token budget.

Score partial. Materialize under a frozen threshold/budget policy.

Hierarchical. Use inherited hierarchical routing/materialization where supported.

Selected Context. Render exactly the selected evidence as freshly encoded ordinary text.

Native contiguous. Encode the selected evidence once as one contiguous native block and reuse it.

Native independent. Reuse independently cached chunks/resources and compose them at request time.

Every condition reports candidate, selected, visible, native, newly materialized, and reused token counts. Following Paper 3.0, partial conditions operate on source-relative logical intervals. Overlap is deduplicated only within one resource domain, so equal numeric offsets in two documents remain distinct native states. In addition to the counts above, each run reports requested tokens before union, unique native tokens, evidence and non-evidence tokens, evidence density, storage shards touched, and cross-shard intervals. Whole-parent and equal-budget wrong-memory conditions act as causal controls where evidence labels permit them. Any fixed budget that omits a required evidence region is reported as a coverage failure, not as evidence that compact materialization is sufficient.

10 Matched Selection and Realization

The most important causal control freezes the selected chunks. For exactly the same selection receipt we compare:

1. fresh selected text;
2. contiguous native reuse;
3. independently cached native fragments at source-local positions;
4. the same independent fragments rebound to packed composition positions;
5. rebound fragments with optional limited contextual repair.

This design ensures that a quality difference after selection cannot be mislabeled as a retrieval failure.

11 Multi-Resource Positional Geometry

11.1 Source and composition coordinates

For resource r and token i, distinguish

$$p_{\mathrm{src}}(r,i) \qquad \text{from} \qquad p_{\mathrm{cmp}}(r,i).$$

The first identifies position in the source's own positional frame. The second is the effective position chosen for the current materialized composition.

A resource-preserving offset can be written

$$p'_{r,i} = b_r + p_{r,i}.$$

The value of b_r is a policy choice, not a consequence of physical K/V residency.

11.2 Required policies

SOURCE_LOCAL. Preserve each resource's source-local coordinates.

GLOBAL_PACKED. Assign selected resources contiguous positions exactly as in freshly packed RAG. This is the primary parity policy.

RESOURCE_ADJACENT. Preserve each resource's internal geometry while independently placing its end near the query. Different resources may overlap in effective RoPE coordinate ranges while remaining separate K/V objects.

RANK_DISTANCE. Map retrieval rank to query distance.

SCORE_DISTANCE. Map a normalized retrieval score to query distance under a frozen mapping.

NON_OVERLAPPING_NEAR_BANDS. Use bounded near-query bands while preserving a monotonic resource ordering.

RANDOM_DISTANCE. Randomize resource offsets under a matched envelope as a negative control.

All policies use one ordinary attention normalization over the concatenated selected K/V. The experiment changes positional transforms, not the definition of attention.

12 Document-Order and Permutation Sensitivity

The minimal experiment compares

$$D_1\,D_2\,Q \qquad \text{with} \qquad D_2\,D_1\,Q.$$

For three or more resources we additionally compare canonical retrieval order, reverse order, and random permutations.

We distinguish two experiments.

Composition sensitivity. Resource identities, selected content, selected-token budget, and model are frozen. Only serialization order and/or positional policy changes. Retrieval is not rerun.

Pipeline sensitivity. Each retrieval/reranking method is allowed to return and pack its natural order. This measures complete RAG behavior.

For permutation set $\mathcal{P}$ we report answer flips, target probability variance, and mean pairwise Jensen–Shannon divergence:

$$S_{\text{order}} = \frac{2}{|\mathcal{P}|(|\mathcal{P}|-1)} \sum_{a<b} JS(P_a \| P_b).$$

A compelling multi-frame result requires maintained or improved task quality in addition to reduced arbitrary permutation variance.

13 Source Distance, Distractors, and Number of Resources

Following the earlier Paper 1.6 plan, source-distance experiments hold evidence fixed while varying its source/global distance from the query. Distractor experiments sweep $0, 2, 4, 8, 16, 32$ irrelevant resources where context permits. Relevant-resource experiments sweep $1, 2, 4, 8$ sources and include redundant, complementary multi-hop, and conflicting evidence.

These controls are especially important for RESOURCE_ADJACENT: placing every resource near the query may reduce lost-in-the-middle effects but could also make irrelevant evidence excessively salient.

14 Positional vs Contextualization Fragmentation

Consider a selected composition A, B, C, Q.

C0: Fresh packed. Freshly encode $[A\,B\,C\,Q]$. This establishes the matched-content reference.

C1: Native source. Reuse independently cached native A, B, C at their source-local effective positions.

C2: Native rebound packed. Keep the independently computed hidden/K/V states but rebind RoPE keys to the positions that A, B, C would occupy in the packed composition.

For pre-RoPE keys,

$$K_{\text{cmp}} = R_{p_{\text{cmp}}} K_{\text{raw}}.$$

For validated post-RoPE storage,

$$K_{\text{cmp}} = R_{p_{\text{cmp}} - p_{\text{src}}} K_{\text{stored}},$$

subject to the exact model's RoPE/scaling implementation.

C3: Rebound plus repair. If positional rebinding does not restore sufficient quality, selectively recompute a fraction of the selected context.

C4: Full recompute. Fresh reprefill provides the full contextualization reference.

A divergence D can summarize the fraction of source-position error recovered by rebinding:

$$R_{\text{pos}} = 1 - \frac{D(P_{\text{fresh}}, P_{\text{rebound}})}{D(P_{\text{fresh}}, P_{\text{source}})}.$$

The residual gap is not automatically purely contextual, but it directly bounds what positional phase correction alone can achieve.

14.1 Is the record gap positional or contextual?

The earlier comparison still changes positional treatment, document contextualization, and cache realization together. We therefore add a causal three-way decomposition after retrieval and ranking have been frozen:

A: FULL_CAUSAL_RAG
Encode the ordinary packed sequence $[D_1\, D_2\, \ldots\, D_n]$ with its standard causal mask.

B: NO_CROSS_DOC_RAG
Keep the identical token sequence, separators, order, and RoPE positions, but let a document token attend only to causal tokens in the same document. Query tokens can still attend every document and earlier query token.

C: PRA_PRE_ROPE_EXACT_PACKED_OFFSETS
Encode each record independently, retain projected keys before RoPE and unchanged values, and apply the exact positions used by B at request time.

P2: SHAPE_MATCHED_PRE_ROPE_REBIND
Repeat B's single packed shape, segmentation, mask, and host-layer walk, but retain pre-RoPE keys and apply host RoPE again at request time. This changes only storage and rebinding, not record-wise projection or attention shapes.

For layer ℓ, record r, and token t, C stores $(K^{\text{pre}}_{\ell rt}, V_{\ell rt})$ and materializes

$$K^{\text{request}}_{\ell rt} = R_\ell(p^{\text{packed}}_{rt}) K^{\text{pre}}_{\ell rt}.$$

The runtime fingerprints the host model revision, layer frequencies, scaling policy, RoPE dimensions, and rotation layout. It refuses materialization under a different contract. The direct implementation captures Qwen's normalized projected key before rotation; it does not infer a raw key by numerically undoing a quantized post-RoPE value.

Let M denote the packed attention mask. For $i \in D_k$,

$$M_{ij} = 1 \iff j \le i \ \wedge \ j \in D_k,$$

whereas for $i \in Q$,

$$M_{ij} = 1 \iff j \in \bigcup_k D_k \quad \text{or} \quad (j \in Q \wedge j \le i).$$

Thus $A - B$ estimates the effect of cross-document causal contextualization, while $B - C$ is the transport and request-time rebinding check. We test the latter at every layer through K/V RMSE and maximum error, then at the output through logits, JS divergence, answer NLL, and generated text. This ordering separates coordinate correctness from representation fidelity, attention-geometry fidelity, and downstream utility.

To test a smaller request-local composition fabric, let $g^K_{\ell i}$ and $g^V_{\ell i}$ be the mean pre-RoPE key and value of selected record i at layer ℓ. The parameter-free baseline uses identity projections:

$$A_\ell = \operatorname{softmax}\Big(G^K_\ell (G^K_\ell)^\top / \sqrt{d_h}\Big), \qquad \widetilde{G}^{K,V}_\ell = G^{K,V}_\ell + A_\ell G^{K,V}_\ell.$$

The default mask is bidirectional all-to-all rather than inherited document order. D appends one contextual K/V gist per selected record in a compact band immediately before the query. E and F instead append request-local copies of the first and last 8 or 32 tokens of each record after adding its contextual gist delta. These are deliberately parameter-free inference baselines. They never overwrite the persistent native K/V and use one host attention softmax over persistent, request-local, and query state. Token-level cross-document edges and gist-level interactions are accounted separately.

Only after the B/C gate do we vary document interaction. The sparse ladder allows the immediately previous document, the top-ranked document, or adjacent boundary windows of 8, 16, 32, and 64 tokens. In the boundary arm, the first w tokens of D_k may attend to the last w tokens of D_{k-1}; later tokens can receive that information through ordinary within-document causality. All arms share the same candidate receipt, BGE-v2-M3 ranking, selected identities, and packed coordinates.

15 Selective Contextual Repair

If C2 remains measurably different from fresh encoding, we test a deliberately simple repair ladder:

$$0\%,\ 5\%,\ 10\%,\ 25\%,\ 50\%,\ 100\%$$

of selected tokens or equivalent compute, together with boundary-window and first-N-tokens-per-resource controls.

A useful result is a Pareto curve:

$$\text{quality/parity recovered} \quad \text{vs} \quad \text{fraction recomputed} \quad \text{vs} \quad \text{TTFT/prefill}.$$

Adaptive repair is postponed until fixed policies establish headroom.

16 Repeated-Query RAG and Persistent Reuse

Real sessions rarely repeat exactly one selected prompt. We therefore construct query sequences such as

$$Q_1 \to \{A, B, C\}, \qquad Q_2 \to \{B, C, D\}, \qquad Q_3 \to \{A, D, E\}.$$

We compare:

1. conventional RAG with full reprefill;
2. conventional engine prefix/KV caching;
3. PRA exact contiguous-block reuse;
4. PRA independent native reuse at source positions;
5. PRA independent native reuse with packed rebinding;
6. packed rebinding plus repair.

We report selection overlap, native chunk-hit fraction, newly encoded and reused tokens, cumulative TTFT, cumulative prefill, wall time, memory residency, official quality, and output flips.

This is the main product-facing experiment: can PRA reuse semantically recurring, non-prefix evidence while an ordinary RAG retriever continues to determine what is relevant?

17 Ordinary Cache Controls

The preliminary warm Native Memory result is not sufficient by itself. On at least one production-style engine we compare:

RAG without cache,
RAG + ordinary prefix/APC cache,
RAG + PRA Native Memory,
RAG + prefix cache + PRA.

Exact-prefix cache hits and query-addressed PRA resource reuse are reported independently.

18 Metrics and Statistics

18.1 Quality

We use official EM/F1/accuracy where defined, gold NLL/log-probability, answer margin, generated sequence agreement, and a frozen semantic judge only where official metrics are insufficient.

18.2 Retrieval

Document Recall@k, supporting-document/span coverage, MRR, nDCG where meaningful, candidate counts, and reranker displacement are reported.

18.3 Context and memory

Metrics include logical candidate tokens, selected/visible/native/new/reused tokens, materialization avoidance, active K/V bytes, and selected/full ratio.

18.4 Composition

We report answer flips, pairwise JS/KL, target-probability variance, source-distance slopes, distractor sensitivity, and fresh/source/rebound divergences.

18.5 Systems

We report index build time, retrieval p50/p95/p99, rerank latency, prefill, TTFT p50/p95/p99, ITL, completion latency, tokens/s, requests/s, memory, K/V bytes, transfer traffic, and cold/warm/hot state.

All matched conditions use paired statistics and bootstrap 95% confidence intervals. Model- and workload-specific sign reversals remain visible rather than being averaged away.

19 Experimental Sequence and Scope

19.1 Phase 0: audit and inheritance

Paper 4.5 artifacts were copied with provenance and labeled preliminary. Dataset, model, retriever, chunker, and generation revisions were verified.

19.2 Phase 1: small-model baseline

Qwen3-1.7B ran the complete controlled and natural mechanism matrices. BM25, FAISS dense, hybrid, and two rerankers were evaluated independently of native realization. Candidate and selection receipts were frozen. The first composed comparison placed fresh selected text against the contiguous native profile, isolating representation and reuse before changing the selector.

19.3 Phase 2: PRA routing/materialization

On identical candidate receipts we ran the routing families and full/partial materialization ladder. The complete RAG+PRA profile matrix crossed conventional selection, PRA second-stage selection, and PRA replacement selection. Realization arms within each selector condition share one immutable selection receipt.

19.4 Phase 3: ordering and position

Seven positional policies and canonical, reverse, and random orders were run on five natural cohorts. Selected resources and distractors were held fixed.

19.5 Phase 4: composition fidelity

Fresh packed, source-native, packed-rebound, and repair conditions isolated positional and contextualization effects.

19.6 Phase 5: real retrieval services

Elasticsearch and Qdrant were deployed remotely. Their embedding, service, and end-to-end client latencies are reported separately. Further vendor breadth is outside this paper's present scope.

19.7 Phase 6: powered 4B study

Qwen3-4B repeated the decisive frozen transport and reduced composition matrix.

19.8 Phase 7: 8B replication

Qwen3-8B repeated the decisive reduced matrix.

19.9 Phase 8: larger/cross-family

Llama-3.1-8B provided the cross-family confirmation. Qwen3-14B was deferred because the 4B/8B/Llama matrix already answered the transport-generalization question and composition remained the limiting mechanism.

19.10 Phase 9: KILT validation

KILT Natural Questions and KILT HotpotQA remain follow-up datasets; adding corpora would not resolve the measured independent-composition failure.

19.11 Phase 10: native records and internal reranking

The final experiment ingests MultiHop-RAG documents through the canonical Paper 4.5 record envelope, resolves explicit leaves or a routed root, and materializes frozen chunk selections. Qwen3-1.7B runs BM25, historical MiniLM, and BGE-v2-M3 selectors on five seeds of 30 questions. Qwen3-4B, Qwen3-8B, and Llama-3.1-8B repeat the MiniLM comparison on one ten-question pilot each.

19.12 Phase 11: pre-RoPE causal decomposition

Five 30-question Qwen3-1.7B cohorts use BM25 candidate generation followed by the pinned BGE-v2-M3 reranker. A/B/C share exact selected records and packed positions. The B/C gate is evaluated before the sparse cross-document mask ladder; reduced 4B, 8B, and Llama replication is conditional on an informative effect rather than automatic model breadth.

19.13 Phase 12: precision and request-local gist composition

The B/C comparison is repeated on a three-question controlled fixture under FP32 and FP16 compute casts of one pinned Qwen3-0.6B source checkpoint and separately converted 8-bit and 4-bit MLX checkpoints. Source precision, runtime weight mode, activation/K/V dtype, quantizer geometry, model revision, and kernel backend are retained separately. P2 is run at every precision. The natural study then freezes BGE-v2-M3 selections for five 30-question Qwen3-1.7B cohorts before evaluating D, E, and F. Controlled precision results are mechanism diagnostics and are not presented as natural-task estimates.

20 Measured Retrieval and Native-Memory Results

20.1 Retrieval remains an independent component

Table 6 reports a frozen 50-question MultiHop-RAG retrieval cohort. The hybrid and stronger reranker results make clear why the native-memory experiment should not silently substitute a weaker selector. The mean reranker timings include first-use costs and are not steady-state serving estimates.

Table 6: Local CUDA retrieval at $K = 20$. "All" requires every supporting document; "mean" averages per-question supporting-document recall.

Method	All support	Mean support	Mean latency (ms)
BM25	.54	.735	7.1
FAISS + BGE dense	.58	.803	12.3
Historical MiniLM reranker	.58	.815	371.9
Hybrid RRF	.70	.870	20.3
BGE-v2-M3 reranker	.78	.897	3868.6

Elasticsearch, Qdrant, and their service-level RRF hybrid reached all-support recall@20 of .56, .52, and .76, respectively. On the RTX 5060 run, mean Elasticsearch service search was 11.46 ms. Qdrant search itself was 2.60 ms; its 23.41 ms end-to-end value included 20.76 ms of BGE query embedding. The hybrid path measured 53.12 ms end to end, including 11.32 ms embedding and 40.75 ms service work. Figure 7 presents the matched local comparison. These results evaluate retrieval, not PRA transport.

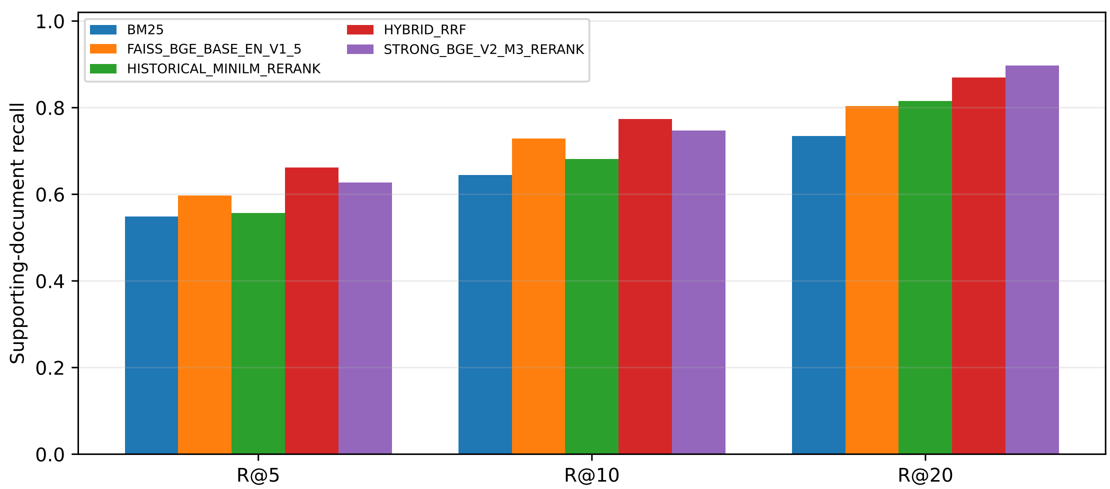

Figure 7: Supporting-document recall for the local retrieval ladder. All methods see the same 50 questions and corpus revision.

20.2 Native record referencing and learned reranking

The earlier composition matrix injects K/V directly to isolate geometry. The record experiment instead exercises the intended user-visible path: ingestion, URI reference, resolver, frozen selection, native chunk materialization, and generation. Explicit leaf references and a routed root produce identical selected chunks and identical outputs in every measured condition; root resolution therefore introduces no additional loss.

Table 7 separates selector improvement from record realization. MiniLM and BGE improve support coverage and packed-text F1 over BM25. The same ordering survives record realization, so learned reranking is useful inside PRA. It does not close the representation gap: all three paired record deltas are negative across all five seeds. BGE reaches the strongest packed F1, .134, and strongest record F1, .080, but its paired delta remains $-.055$ (95% CI $[-.076, -.035]$). Gold-answer NLL moves in the opposite direction for the 1.7B record conditions (for BGE, 5.820 to 2.827), while generated-answer F1 falls. We therefore report both and do not treat teacher-forced NLL alone as evidence of better answering.

Table 7: Five-seed native-record experiment on Qwen3-1.7B (150 questions). Support is mean supporting-document coverage. Latency is reranker latency and includes first-use costs. Intervals are seed-bootstrap 95% CIs.

Selector	Support	Packed F1	Record F1	Paired Δ F1	Rerank (ms)
BM25	.375	.093	.059	$-.034$ $[-.044, -.028]$	0.0
MiniLM	.558	.124	.079	$-.045$ $[-.082, -.021]$	497.8
BGE-v2-M3	.638	.134	.080	$-.055$ $[-.076, -.035]$	4788.5

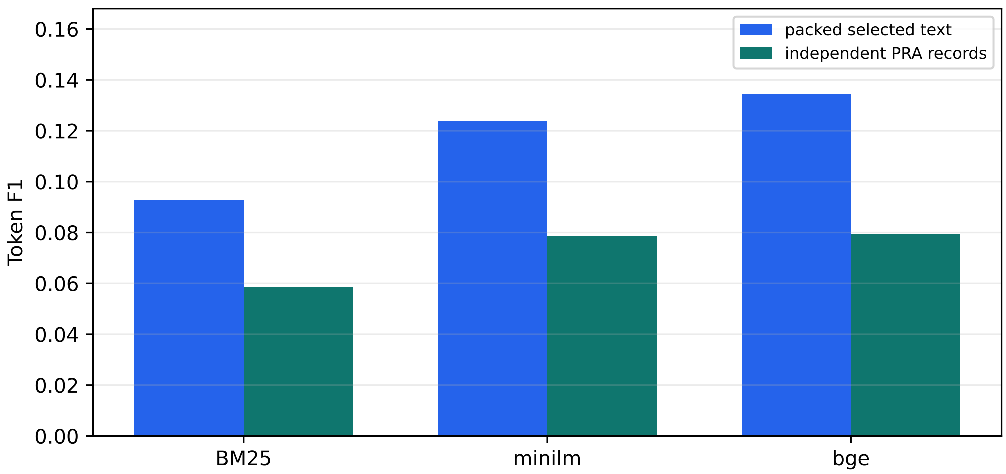

Figure 8: Better reranking improves both packed and record-relative answering, but does not remove the independent-record quality gap. All bars use the same frozen selection within each selector.

Record-relative execution is much less sensitive to arbitrary record order. Across canonical, reverse, and two random permutations, mean pairwise first-step JS falls from .04595 for packed text to .000647 for records (98.6% lower), and unique outputs per question fall from 2.68 to 1.65. Mean F1 variance across orders falls from .00437 to .00107. Figure 9 shows the two direct robustness measures. This is a mechanistic result, not a quality win: the canonical record F1 remains below packed F1.

The changing-query record test contains 25 four-turn sequences. Exact-prefix matching reuses 19.1 tokens per turn on average, mostly prompt scaffolding; stable chunk identities expose 327.7 reusable native token states while 1693.5 new states are encoded. Packed F1 is .1214 and record F1 is .0775, a paired delta of $-.0438$ (95% CI $[-.0733, -.0213]$). The record path thus provides the desired non-prefix addressability, but the original record-reference hypothesis is not validated at 1.7B because task quality is not preserved.

The initially positive Qwen3-4B seed was repeated over all five predeclared seeds. Across 50 questions, packed F1 is .133 and record F1 is .112, a seed-level mean delta of $-.021$ (bootstrap 95% CI $[-.055, .007]$). The seed deltas are $+.018, +.006, -.024, -.089, -.015$: the apparent seed-11

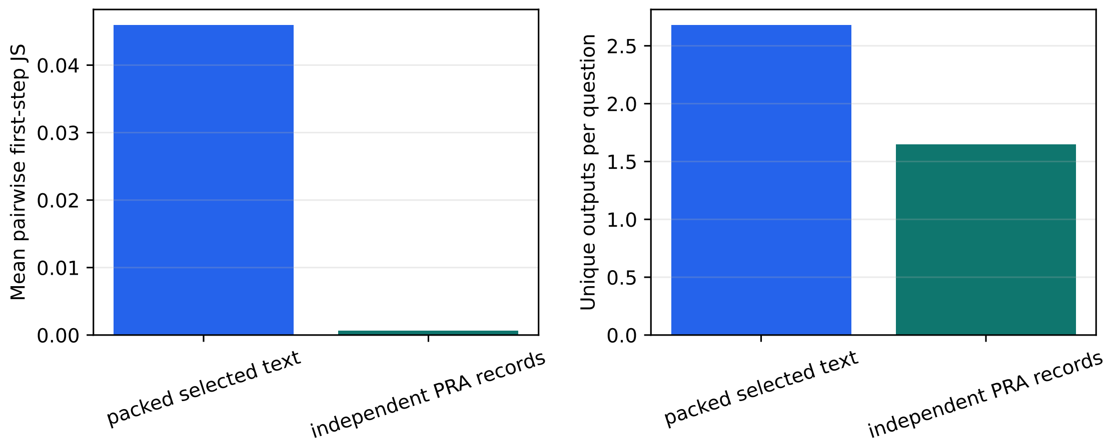

Figure 9: Independent records reduce sensitivity to arbitrary cross-document serialization. The lower variance is interpreted together with, not instead of, the quality regression in Table 7.

gain does not replicate. Mean answer NLL changes from 3.076 to 2.756, but its delta interval also crosses zero ($-.320$, $[-.697, .140]$), and output agreement remains zero. The one-seed Qwen3-8B and Llama-3.1-8B pilots are also negative. Scale alone therefore does not qualify independent records as a runtime default.

Table 8: Native-record scale/family checks with MiniLM selection. Qwen3-4B uses five seeds and 50 questions; the 8B rows remain seed-11 ten-question pilots. P/R are packed/record values.

Model	F1 P/R	ΔF1	NLL P/R	JS
Qwen3-4B	.133/.112	−.021	3.076/2.756	.185
Qwen3-8B	.182/.115	−.067	2.356/2.375	.099
Llama-3.1-8B	.173/.162	−.011	.961/1.679	.130

20.3 Pre-RoPE rebinding localizes, but does not erase, the record gap

The final causal decomposition contains 150 MultiHop-RAG questions: 30 in each of five independently selected seed cohorts. Candidate generation, BGE-v2-M3 ranking, selected identities, separators, packed token order, and request positions are frozen within each A/B/C triple. The mean selected prefix contains 1330 token states and occupies 145.4 MiB across the model's native K/V layers.

Table 9 gives the task results. Full causal packing (A) reaches .119 token F1 and .467 official score, compared with .110 and .400 when document-to-document attention is removed (B). The seed-level A–B F1 difference is +.0084 with a bootstrap 95% interval $[-.0171, .0292]$ and exact two-sided sign-flip $p = .50$. Gold-answer NLL points in the opposite direction: A is worse than B by 2.455, with interval $[1.802, 3.067]$. All five NLL effects have the same sign, but five seeds give the minimum exact two-sided $p = .0625$. Cross-document contextualization therefore changes model behavior materially, but this cohort does not support the simpler claim that it improves every answer metric.

B and C are much closer at task level. Pre-RoPE records at exact packed positions (C) reach .108 F1 and 3.474 NLL, versus .110 and 3.486 for blocked packing. The seed-level B–C F1 difference

Table 9: Five-seed pre-RoPE causal decomposition (150 questions). All conditions share frozen retrieval and packed positions. Official is the MultiHop-RAG token-intersection score. Edges count allowed document-to-document attention pairs; query visibility is unchanged.

Condition	Token F1	Gold NLL	Official	Cross-doc edges
A full	0.119	5.940	0.467	665894
B isolated	0.110	3.486	0.400	0
C pre-RoPE	0.108	3.474	0.393	0
previous	0.111	6.167	0.440	333364
top-ranked	0.115	6.131	0.473	334414
boundary 8	0.118	6.144	0.447	192
boundary 16	0.107	6.110	0.393	768
boundary 32	0.111	6.083	0.407	3072
boundary 64	0.106	6.056	0.427	12285

is $+.0022$ ($[-.0019, .0063]$, $p = .4375$); signed NLL differs by $+.0119$ ($[.0052, .0200]$, $p = .0625$), and mean absolute NLL error is .0701. Generated outputs agree in 101/150 cases and mean first-step JS is 8.80×10^{-4}. These measurements support close downstream agreement, not numerical equivalence: no first-step logit hash matches exactly.

Table 10: Blocked packed encoding (B) versus independently encoded pre-RoPE records rebound to B's coordinates (C). Worst-layer values are maxima over all five runs.

B/C diagnostic	Value
Generated-output agreement	101/150
Exact first-step-logit hashes	0/150
Mean first-step JS	0.000880
Mean $\lvert\Delta\text{NLL}\rvert$	0.0701
Worst layer key RMSE	0.0767
Worst layer value RMSE	0.5608

The source of this distinction is observable. A historical tiny unquantized Qwen fixture matched B/C within 2×10^{-5} under its original runtime, but the explicit sweep below shows that this tolerance is kernel-version and shape dependent. Under the production 4-bit checkpoint, layer-0 mean value RMSE is 3.10×10^{-5}, but mean value RMSE grows to .274 by layer 27. Independent and packed projection/attention shapes can take different quantized kernel paths; small early differences are amplified through the decoder. Figure 10 reports mean and p95 rather than hiding this behavior behind either an exact tiny fixture or one worst-case maximum. The binding algebra and host-frequency contract pass, while exact 4-bit numerical parity remains an open runtime qualification gate.

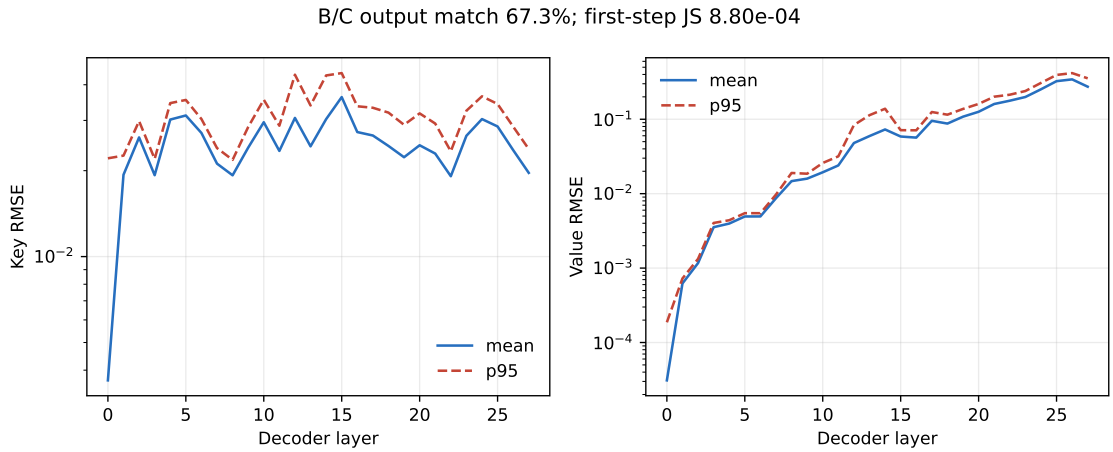

Figure 10: Layerwise B/C K/V divergence for 150 paired 4-bit executions. Values are measured after request-time rebinding to identical packed positions.

20.4 Precision qualification separates rebinding from shape effects

The new controlled precision sweep uses three matched questions at every precision. FP32 here means FP32 execution after casting the pinned BF16 source checkpoint; it does not recreate unavailable FP32 source weights. The INT8 and INT4 rows use separately published groupwise-converted checkpoints from the same source family, so differences between those two rows combine weight conversion and kernel effects and are not a bitwise quantizer ablation.

Table 11: Controlled B/C precision qualification on Qwen3-0.6B (three matched questions per row). K/V are retained at the runtime activation dtype.

Precision	Pairs	Output match	JS	$\lvert\Delta\text{NLL}\rvert$	Max K RMSE	Max V RMSE
FP32	3	1.000	0.000004	0.0009	0.0018	0.0065
FP16	3	1.000	0.000009	0.0036	0.0049	0.0194
INT8	3	1.000	0.000621	0.0500	0.1161	0.2003
INT4	3	1.000	0.001082	0.0083	0.1005	0.2130

The precision ordering is clear in the distributional metric: mean first-step JS rises from 3.91×10^{-6} in FP32 to 8.74×10^{-6} in FP16, 6.21×10^{-4} in INT8, and 1.08×10^{-3} in INT4. Output agreement remains 3/3 in every row, but no B/C logit hash is exact. Worst-layer value RMSE rises from .0065 to .0194, .2003, and .2130. NLL and key RMSE are not strictly monotonic: INT8 has larger mean absolute NLL drift than INT4 (.0500 versus .0083), while their worst key RMSE reverses (.1161 versus .1005). We therefore accept the narrower hypothesis that lower precision substantially amplifies independent-shape drift, not a universal monotonic law for every downstream metric.

P2 makes the source of the residual much sharper. Holding the packed shape and mask fixed while changing only post-RoPE storage to pre-RoPE storage plus request-time host rebinding gives zero K/V error, zero JS, zero NLL error, and exact output/logit agreement for all 12 comparisons.

Thus the host RoPE transport is exact in the measured path. The nonzero B/C residual comes from separately shaped record executions and their accumulated finite-precision operation order, not from an incorrect rebinding identity. Figure 11 shows the associated layerwise amplification.

Table 12: P2 shape-matched storage/rebinding control. The same packed host execution is used on both sides; only pre-RoPE persistence and rebinding differ.

Precision	Pairs	Output match	JS	$\lvert\Delta\text{NLL}\rvert$
FP32	3	1.000	0.00000000	0.000000
FP16	3	1.000	0.00000000	0.000000
INT8	3	1.000	0.00000000	0.000000
INT4	3	1.000	0.00000000	0.000000

The sample is intentionally small; its role is causal localization rather than confidence about natural answer quality.

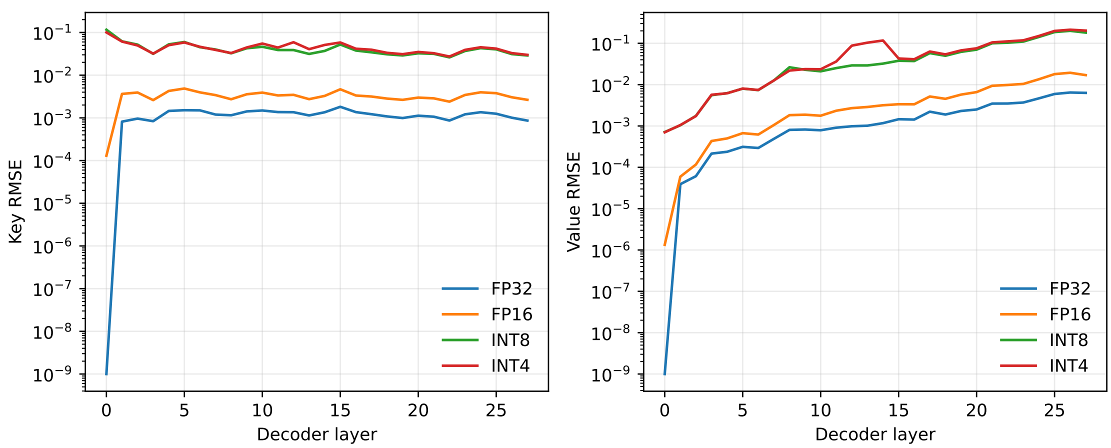

Figure 11: Layerwise independent-record B/C error under the explicit precision ladder. P2 is exactly zero at every layer and is omitted from the log axis.

20.5 Parameter-free gist composition fails the natural-task gate

The main composition test returns to the five predeclared 30-question MultiHop-RAG seed cohorts. Retrieval and BGE-v2-M3 ranking are frozen once per question; A/B/C, P2, and the three request-local composition arms see the same selected records, order, token budget, and query. D appends four contextual gist tokens on average. E and F append conditioned copies of each record's first and last 8 or 32 tokens, respectively. Persistent record K/V is never modified.

Table 13: Five-seed parameter-free cross-document composition (150 natural questions). Interactions combine the 16-edge all-to-all gist graph with boundary-conditioned token edges; they are not dense document-attention edges.

Condition	Token F1	Official	Gold NLL	ΔF1 vs C	Interactions
C pre-RoPE	0.108	0.393	3.474	+0.000	0
D gist append	0.007	0.020	8.135	-0.101	16
E boundary 8	0.000	0.007	11.173	-0.108	80
F boundary 32	0.000	0.007	11.383	-0.108	272

The gate closes decisively. Independent pre-RoPE C reaches .108 F1, .393 official score, and

3.474 NLL. Gist append D falls to .007 F1 and .020 official score while NLL rises to 8.135. Both boundary-copy variants have zero mean F1, .007 official score, and NLL above 11. The paired D–C F1 change is $-.1007$ with a seed-bootstrap 95% interval $[-.1257, -.0691]$; E–C and F–C are both $-.1081$ with interval $[-.1274, -.0838]$. Every seed effect has the same sign; with five seeds the exact two-sided sign-flip value is nevertheless bounded at $p = .0625$. NLL increases by 4.661, 7.699, and 7.909 for D, E, and F.

The mechanisms are computationally small but semantically wrong. D performs 16 gist interactions and adds four ephemeral K/V tokens in .44 MiB. E and F use 80 and 272 total request interactions, add 64 and 256 ephemeral tokens, and occupy 7.0 and 28.0 MiB. Measured composition time is 10.94, 13.09, and 14.53 ms. These figures include one layerwise identity-projection gist graph and boundary realization, but exclude persistent source encoding already accounted elsewhere.

Table 14: Request-local composition cost on the Apple M4 Pro. FLOPs are an operation-count estimate; latency and bytes are measured.

Condition	Gist edges	Boundary edges	Ephemeral tok.	MiB	Comp. ms	GFLOP
D gist append	16	0	4	0.4	10.94	0.002
E boundary 8	16	64	64	7.0	13.09	0.006
F boundary 32	16	256	256	28.0	14.53	0.017

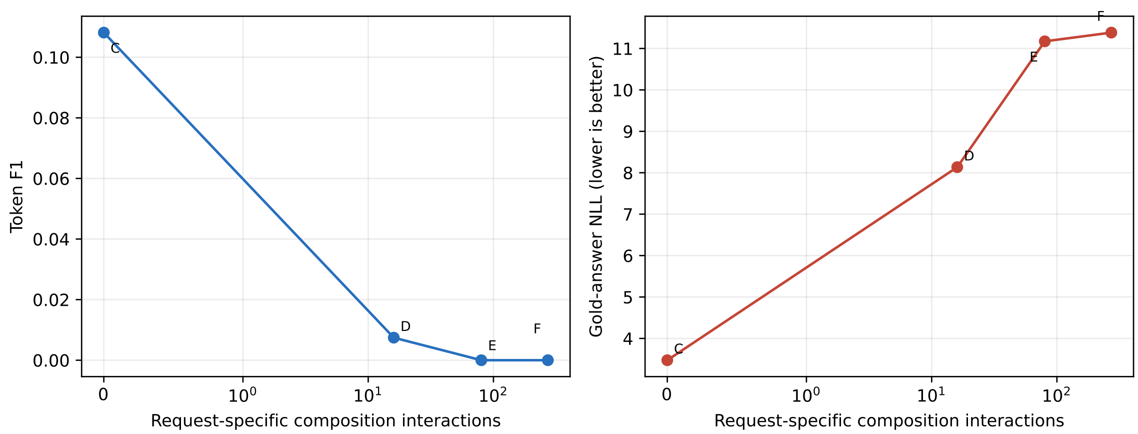

Figure 12: Parameter-free composition quality against request-specific interaction count. A tiny interaction graph is not useful when its K/V states are not calibrated for the frozen decoder.

The failure distinguishes two superficially similar sparse mechanisms. The earlier boundary-8 *attention mask* lets document hidden states acquire cross-document information while those states are computed and reaches .118 F1. E instead inserts post-hoc boundary K/V copies conditioned by means that the base model was never trained to consume, and reaches zero. The likely mechanism is off-manifold K/V plus attention-normalization competition, though the experiment identifies the intervention rather than proving that internal explanation. Increasing boundary width only worsens NLL. We therefore do not implement the optional low-rank correction in this inference-only iteration: the next gate requires a trained composition projection with packed-RAG distillation and held-out task validation.

Request-time rotation itself is inexpensive in this implementation: .69 ms on average. B and C have nearly identical generation-only TTFT (68.1 and 68.3 ms) and total generation latency (177.9

and 177.5 ms). Independent source encoding is still paid during the cold path and is not claimed as free; Table 15 keeps encode, binding, and serving time separate.

Table 15: Mean Apple M4 Pro mechanism timings. Encode is cold source encoding; TTFT and total are measured after native memory is available.

Condition	Native tok.	MiB	Encode ms	Bind ms	TTFT ms	Total ms
A full	1330	145.4	611.1	0.00	88.9	197.7
B isolated	1330	145.4	657.1	0.00	68.1	177.9
C pre-RoPE	1330	145.4	649.0	0.69	68.3	177.5

The sparse interaction ladder does not yield a monotonic quality law. An eight-token boundary window permits only 192 cross-document edges on average, 99.97% fewer than A's 665,894, while reaching .118 F1 versus A's .119. Yet its official score is lower (.447 versus .467) and its NLL is worse (6.144 versus 5.940). The top-ranked-to-all policy slightly exceeds A on the official score (.473) but is lower on F1 and worse on NLL. Confidence intervals overlap broadly. Figure 13 shows task quality and NLL together; Figure 14 makes the absence of a smooth interaction-budget frontier visible. Boundary interaction is a promising composition primitive, but no tested fixed mask is qualified as a replacement for full packing.

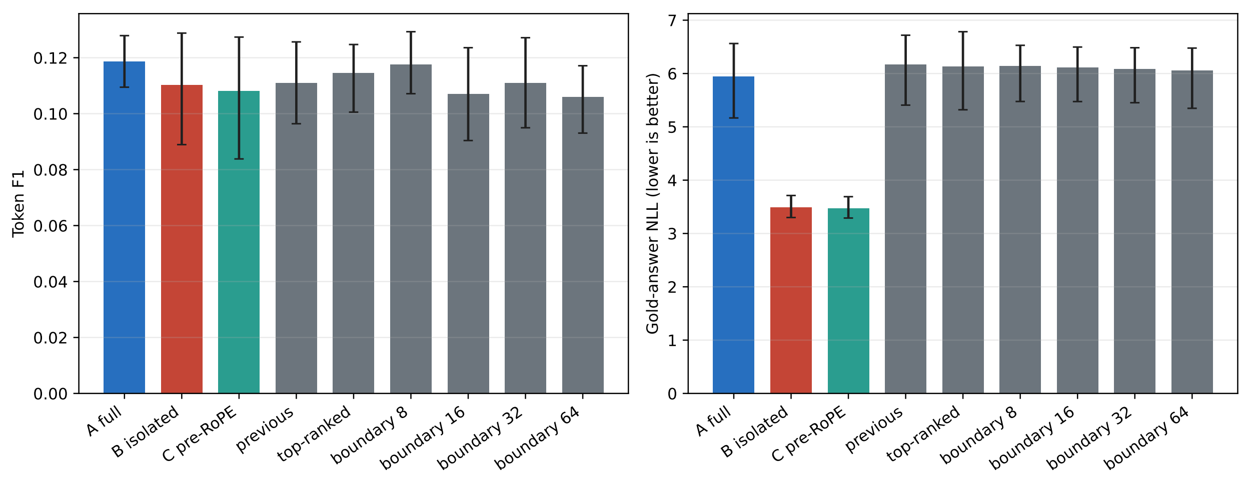

Figure 13: Task F1 and gold-answer NLL for A/B/C and the sparse mask ladder. Intervals bootstrap five seed-cohort means.

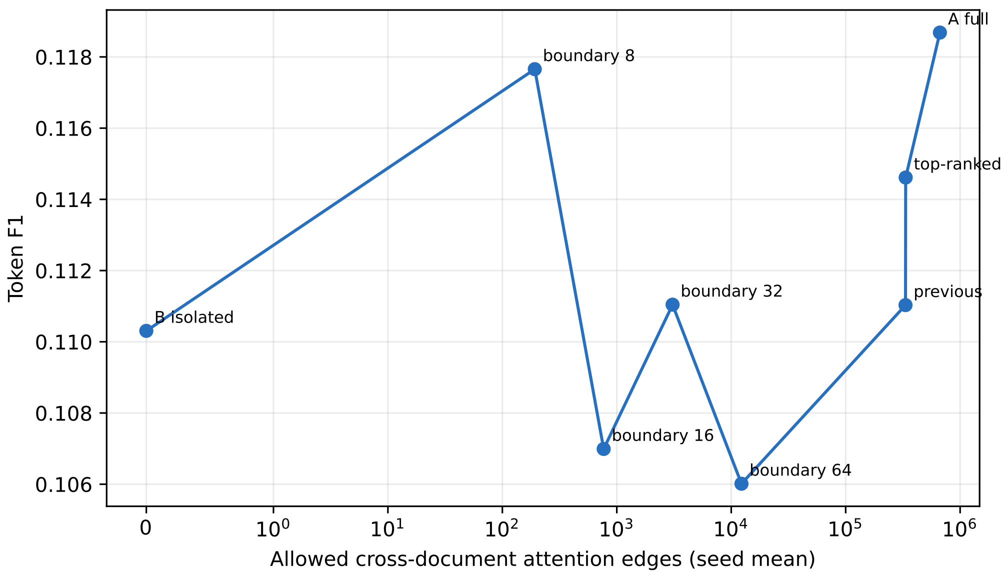

Figure 14: Token F1 against allowed document-to-document attention edges. More interaction is not monotonically better on the measured cohort.

20.6 Reduced model and family replication

Table 16 adds preregistered ten-question checks at 4B and 8B scale. These are mechanism replications, not powered substitutes for the five-seed 1.7B result. B and C remain behaviorally close in every row. Their F1 is identical for Qwen3-8B and Llama-3.1-8B; generated outputs agree in 8/10, 10/10, and 10/10 cases for Qwen3-4B, Qwen3-8B, and Llama-3.1-8B, respectively. No row has an exact first-step-logit hash match, so the table does not relabel behavioral agreement as numerical identity.

Table 16: Pre-RoPE reduced scale and family check. A/B/C denote full packed, document-isolated packed, and independently encoded pre-RoPE records at exact packed positions. Only Qwen3-1.7B uses five independent seed cohorts.

Model	Cohort	A/B/C F1	B/C output	JS	$\|\Delta\text{NLL}\|$	Worst V RMSE	Bind ms
Qwen3-1.7B	5×30	.119/.110/.108	101/150	8.80×10^{-4}	.0701	.5608	.69
Qwen3-4B	1×10	.183/.183/.181	8/10	4.05×10^{-4}	.0313	.1678	.88
Qwen3-8B	1×10	.187/.169/.169	10/10	3.14×10^{-4}	.0438	.1340	.87
Llama-3.1-8B	1×10	.196/.293/.293	10/10	1.41×10^{-6}	.0012	.0014	3.75

The numerical residual is architecture-dependent. Qwen's worst-layer value RMSE falls from .168 at 4B to .134 at 8B, whereas Llama's is .0014 and its first-step JS is 1.41×10^{-6}. This is useful cross-family evidence for the exact host-frequency implementation, including Llama's piecewise RoPE, but three one-seed pilots do not define a model-size scaling law. Full causal packing is also not uniformly better: A exceeds B on Qwen3-8B F1, is nearly tied at 4B, and is lower on this small Llama cohort. Request-time key binding costs below 1 ms for both Qwen pilots and 3.75 ms for Llama. The safe conclusion is therefore narrow: pre-RoPE C closely tracks its causally isolated B target across two model families, while the remaining finite-precision residual and the utility of A's cross-document interaction depend on model and cohort.

20.7 Natural composition and contextual repair

We next freeze the historical cross-encoder's selection for five independently sampled cohorts of ten natural questions, using four resource orders per question. Table 17 compares each native realization with fresh packed encoding. Contiguous reuse is numerically exact in all 200 comparisons. Independent source-local encoding is not: changing only the RoPE phase to the fresh packed coordinates improves mean JS and NLL error but does not restore the missing cross-resource causal context. At 50% recomputation, repairing a contiguous prefix of later resources is more faithful than repairing boundaries, recovering 123 of 200 outputs versus 120. Only full recomputation reaches exact parity in all five cohorts.

Table 17: Five-seed natural independent-composition experiment (200 order-specific comparisons). NLL is mean absolute gold-answer NLL error versus fresh packed.

Realization	Output parity	JS	$\lvert\Delta\text{NLL}\rvert$
Contiguous native block	200/200	.000	.000
Independent source-local	17/200	.168	2.638
Global-position rebound	21/200	.138	2.305
Boundary repair, 25%	92/200	.040	.782
Boundary repair, 50%	120/200	.026	.577
Later-resource prefix repair, 50%	123/200	.014	.417
Full repair	200/200	.000	.000

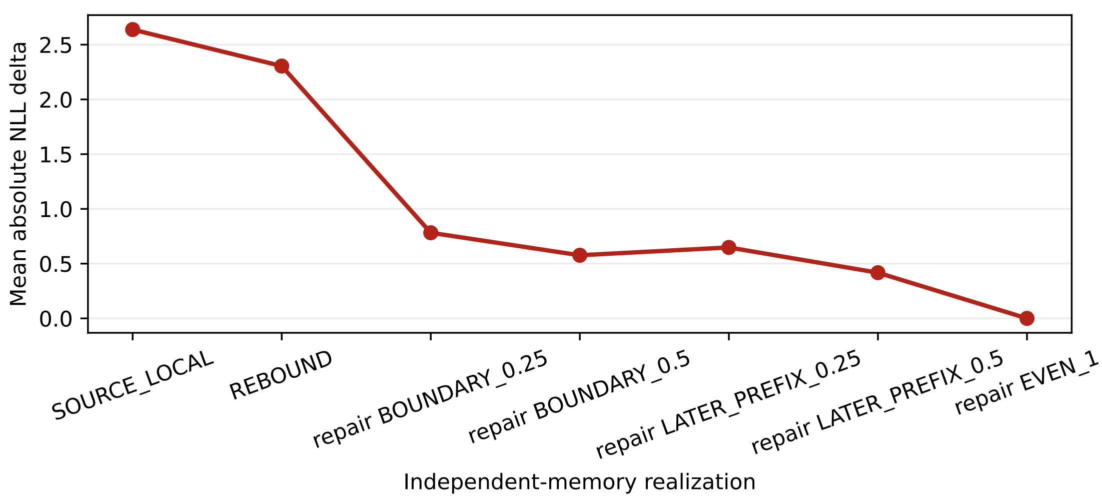

Figure 15: Position rebinding and deterministic contextual repair reduce the composition error, but all five cohorts reach exact parity only at full repair. Repair timings are diagnostic because fresh packed states were computed first.

Serialization itself is consequential: 39 of 50 questions generated more than one output across canonical, reverse, and two random orders. This establishes order sensitivity but does not identify an alternative position policy that preserves quality.

20.8 Alternative positional geometries

A five-seed sweep evaluates the Paper 1.6 position policies on the same 50 questions and four orders per question. None dominates packed-position rebinding. GLOBAL_PACKED gives the lowest mean JS divergence (.138) and absolute NLL error (2.305) among the reusable geometries, preserving 21 of 200 outputs. Non-overlapping near bands preserve 23 outputs, but increase JS to .149 and NLL error to 2.367. Rank, score, and random distance preserve 19–20 outputs with JS .151–.153. RESOURCE_ADJACENT is weakest: 13 output matches, .197 JS, and 2.807 NLL error. Position geometry changes the failure, but does not replace missing cross-resource contextualization.

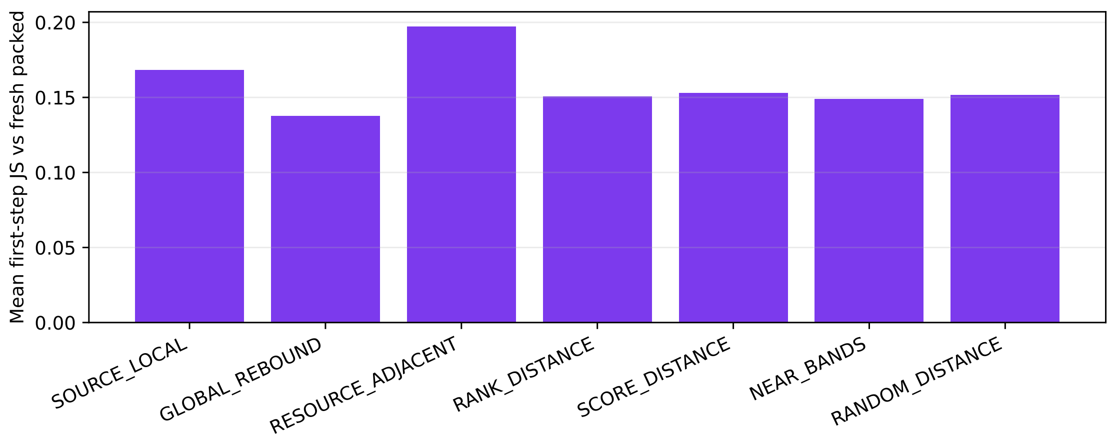

Figure 16: First-step divergence from fresh packed encoding under alternative position policies (200 order-specific comparisons per policy across five cohorts).

20.9 Held-out repair calibration

The fixed ladder makes a small adaptive test meaningful. We fit a bounded lookup policy on seeds 11, 23, and 37, then freeze it for seeds 71 and 101. The policy observes only question type, selected native-token length, and resource count. Within each coarse feature cell it chooses among rebound and the measured repair actions up to 50%, minimizing first-step JS plus a linear recomputation penalty. It never observes answers, support labels, or evaluation outcomes at request time.

Table 18 reports 80 held-out order-specific cases. The policy chooses 25% repair for 28 cases and 50% for 52. It recovers 66.25% of fresh outputs, compared with 67.5% for the calibration-selected fixed 50% action, while repairing 553 rather than 658 tokens per case (16.0% less). This is a modest fidelity–cost improvement, not an answer-quality improvement: policy F1 is .121 versus .148 for fresh packed encoding and .170 for unrepaired rebound. First-step imitation is therefore an insufficient objective for promoting adaptive composition.

20.10 Partial materialization

Table 19 holds selected resources and order fixed, then reduces active native K/V across the five cohorts. Score-ranked 75% materialization lowers active K/V from 1320.4 to 990.3 and TTFT from 88.2 to 68.6 ms, but mean token F1 falls from .234 to .181 and only 19 of 50 outputs match full materialization. Smaller fractions are non-monotonic. The equal-budget evidence oracle improves support coverage over score selection at 25% (.393 versus .318) but does not recover full-context F1.

Table 18: Held-out repair calibration on seeds 71 and 101. Recovery is exact generated-output agreement with fresh packed encoding.

Condition	Recovery	JS	Repaired tokens	F1
Fresh packed reference	1.000	.0000	0	.148
Unrepaired rebound	.100	.1517	0	.170
Best fixed (later-prefix 50%)	.675	.0139	658	.121
Query-conditioned policy	.663	.0177	553	.121

The wrong-memory control reduces support coverage to .132. A fixed 75% policy therefore fails the replication gate.

Table 19: Canonical-order partial-materialization diagnostic.

Policy	Active K/V	Support	F1	NLL	TTFT (ms)
Full	1320.4	.432	.234	5.180	88.2
Score, 75%	990.3	.417	.181	4.176	68.6
Score, 50%	660.2	.368	.117	4.111	66.1
Score, 25%	330.1	.318	.137	4.177	62.1
Score, 12.5%	165.1	.298	.098	4.927	61.8
Evidence oracle, 25%	330.1	.393	.150	4.350	62.9
Wrong memory, 25%	330.1	.132	.139	4.797	62.1

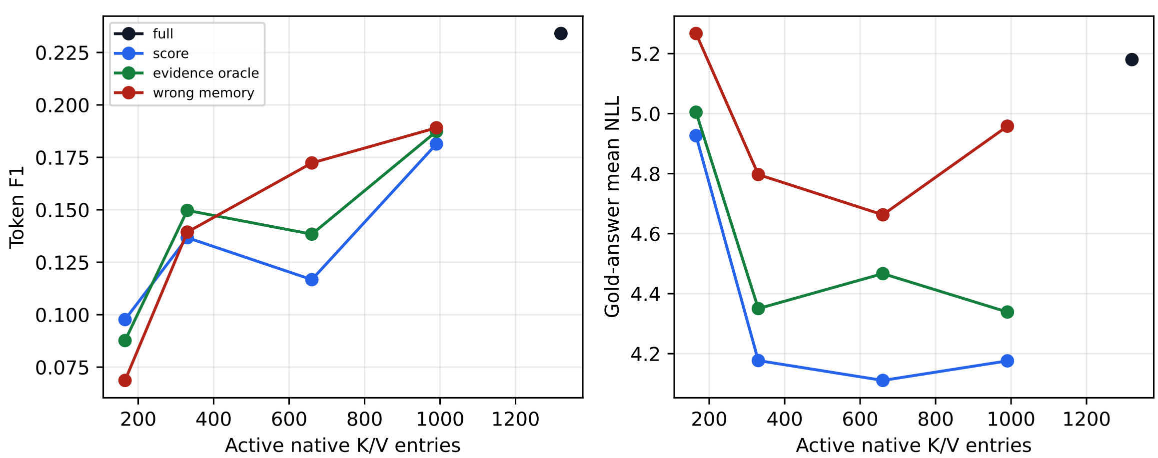

Figure 17: Quality and gold-answer NLL against active native K/V. The different metrics need not rank small-sample generations identically.

20.11 Changing-selection reuse

The product-relevant test groups 100 natural questions across five seeds into 25 four-turn sequences chosen to overlap selected resources. Ordinary exact-prefix caching reuses zero evidence tokens because each question changes the serialization. Resource-addressed PRA reuses 372.1 of 1325.0 selected K/V entries per turn on average (28.1%), reducing newly encoded entries by the same fraction. Measured total time falls from 716.6 to 557.6 ms (22.2%).

This is not an end-to-end quality win yet. Global rebound preserves 7% of fresh outputs (seed-bootstrap 95% interval 3–10%) and reaches .172 mean token F1 versus .217 for fresh packed encoding. A 25% diagnostic repair raises output parity to 44% (33–55%) and F1 to .179. Once repaired-token work is counted, however, it requires 1284.1 newly encoded or recomputed token states per turn, only 3.1% below fresh packed encoding. Its measured copy latency is not reported as deployable repair latency because the fresh states already exist in this diagnostic. Partial 50% materialization lowers newly encoded K/V by 57.1% and measured total time by 47.2%, but preserves only 6% of outputs and reaches .150 F1. PRA therefore finds reuse that an exact prefix cache cannot express, while the present independently encoded representation still needs a better contextualization mechanism.

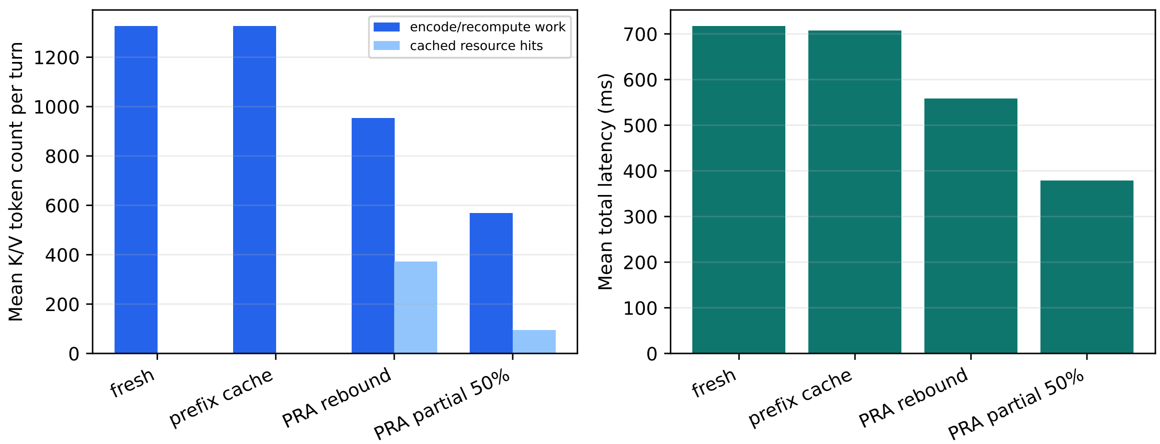

Figure 18: Five-seed mean per-turn K/V accounting and measured latency under changing selections. Encoding/recomputation and cache hits are shown as separate bars rather than as an active-context stack. Diagnostic repair is omitted because copying precomputed fresh states is not a deployable timing.

20.12 Model scale and family replication

We repeat the selector-frozen contiguous transport control on one 20-question cohort at 4B and 8B scale and on a second model family. All 120 cold/warm pairs match generated output, first-step-logit hash, and gold-answer NLL exactly. Table 20 reports warm timing. Native TTFT is 5.9–6.5% of selected-text re-prefill, but remains 4.6–7.2% slower than an exact prefix hit. The Llama row shows that the result is not specific to Qwen's tokenizer or model implementation. The quality differences among rows are model effects, not transport effects, because text and native realization agree within every row.

Table 20: Reduced MultiHop-RAG scale replication (20 questions, seed 11). TTFT columns are selected-text re-prefill, warm native, and exact prefix.

Model	F1	NLL	Pair parity	Text/native/prefix TTFT (ms)
Qwen3-4B	.290	2.290	40/40	2572 / 167 / 156
Qwen3-8B	.308	2.100	40/40	4724 / 280 / 268
Llama-3.1-8B	.365	1.188	40/40	4573 / 274 / 261

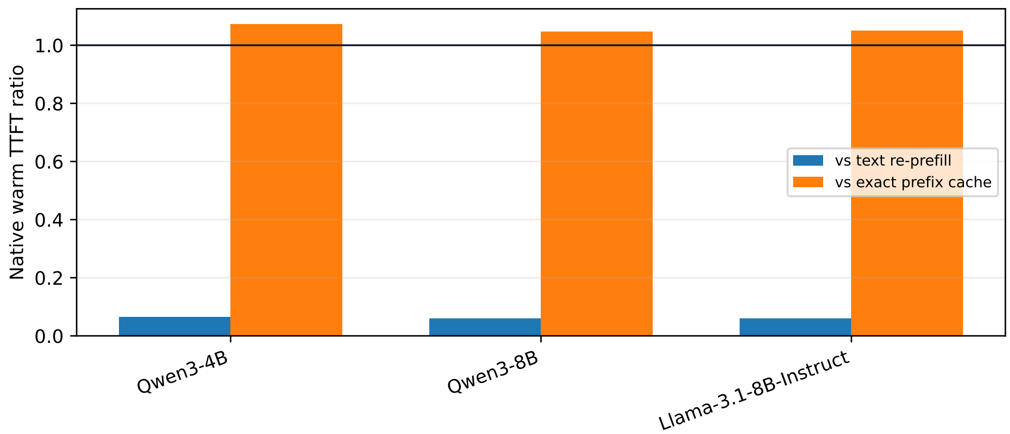

Figure 19: Warm native TTFT relative to text re-prefill and exact-prefix reuse. The horizontal line marks equal cost; lower is better.

The reduced composition matrix is less uniform. On the same five questions in canonical and reverse order, 50% later-resource prefix repair recovers 8/10 Qwen3-4B outputs, 5/10 Qwen3-8B outputs, and 7/10 Llama-3.1-8B outputs. Packed rebinding alone recovers 2/10, 0/10, and 3/10, respectively; full repair is 10/10 for every model. Mean JS after 50% repair is .00039, .00200, and .00143. This suggests useful bounded repair at larger scale, but neither a monotonic model-size law nor parity.

The Llama experiment also exposed an implementation-sensitive requirement. Llama-3.1 uses piecewise long-context RoPE frequencies rather than a single base. An initial single-base transform worsened JS to .526. The corrected rebinder reads the host layer's exact frequency tensor; only corrected values are reported above. This is why cross-family validation is necessary even when contiguous native transport already passes.

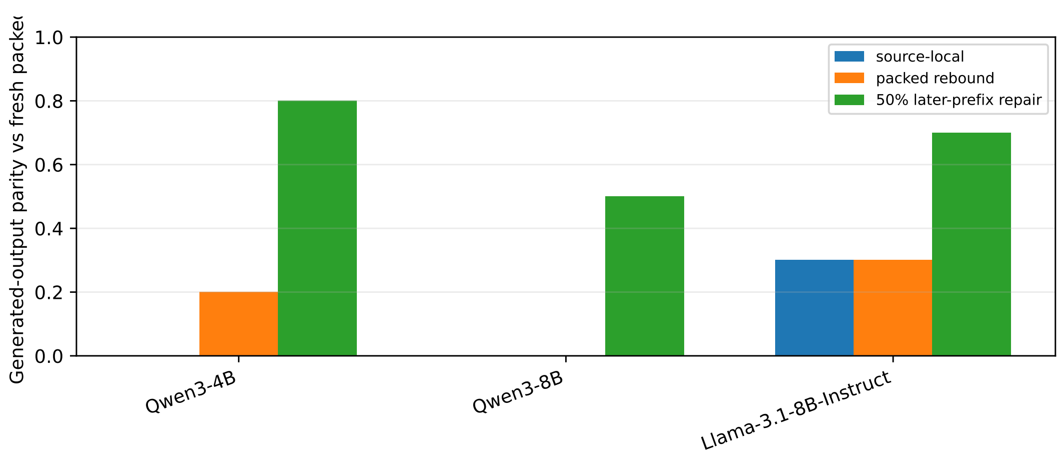

Figure 20: Independent-composition output parity on the reduced scale cohort. Full repair, omitted for scale, is 1.0 for all three models.

21 Interpretation and Falsification

Three claims now have different evidential status. First, contiguous native realization is established for the measured model and cohorts: selection, output, logits, and answer NLL agree. Second, non-prefix resource reuse is measurable where exact-prefix reuse is zero. Third, useful composition is not yet established: RoPE rebinding removes only part of the error, and the tested bounded repair policies do not recover fresh behavior reliably.

The A/B/C decomposition identifies two separable causes. Denying packed documents cross-document attention changes the answer distribution, although its direction is metric-dependent on this cohort. Once that interaction is removed, exact packed-position pre-RoPE rebinding closely approximates the blocked computation at task level. This localizes most of the conceptual gap in cross-resource contextualization. It does not certify bitwise transport through a quantized runtime: B/C output agreement is 67.3%, and small early-layer differences grow through the 4-bit decoder even though the same implementation passes an unquantized all-layer fixture. Rebinding and numerical-runtime qualification are therefore separate gates.

The native-record experiment sharpens the third conclusion. The failure is not an artifact of bypassing the public abstraction: canonical Paper 4.5 records, explicit references, rooted routing, and strong rerankers reach the same boundary. Better selectors improve evidence coverage and downstream quality, but frozen-selection record realization regresses on both the powered 1.7B comparison and the replicated 4B comparison. Conversely, explicit and rooted references agree exactly, so the reference parser and root resolver are not the source of the measured loss. The large reduction in order sensitivity is real, but lower sensitivity is useful only if subsequent training or composition repair also preserves answer quality.

The results therefore falsify the strongest form of the composition hypothesis. Post-RoPE K/V is not a freely movable document representation. Later-layer states carry causal context from the serialization in which they were formed. This does not invalidate persistent native memory, but it narrows its present safe use to contiguous blocks, unchanged selections, or explicitly repaired fragments.

The sparse-mask ladder also rejects a simple "more interaction is better" account. An eight-token boundary window uses 99.97% fewer document-to-document edges than full packing and nearly matches its mean F1, yet loses on official score and NLL; wider windows are non-monotonic. A learned or query-conditioned interaction policy remains plausible, but the present fixed policies do not pass a quality-preserving promotion gate.

Partial materialization remains an open adaptive-control problem. The 75% point matches mean F1 on the first seed but not across five cohorts; non-monotonic smaller budgets and output flips show that evidence coverage and attention competition cannot be reduced to token count alone. The held-out query-calibrated repair policy saves 16% of repair work relative to its best fixed comparator at nearly the same output-recovery rate, but lowers answer F1 and therefore fails the product-quality gate. A learned policy needs an answer-aware training objective and a new held-out validation before becoming a product profile. Likewise, the measured order sensitivity motivates multi-frame geometry but does not count as evidence that any alternative geometry is better.

The strongest baseline boundary is also retained: for an identical repeated serialized prefix, ordinary prefix caching is faster than native PRA. PRA's distinct opportunity begins when resources recur across changed questions, orders, or evidence combinations. Even there, the current cost saving becomes useful only if representation fidelity is repaired.

22 Systems and Reproducibility

Every run persists dataset/corpus revision, query identity, candidate receipt, candidate scores, reranker scores, selected resource/chunk IDs, selected order, spans, materialization receipt, source and composition positions, position policy, model/tokenizer revision, generation settings, backend/index version, hardware, and run/commit IDs.

Real retrieval services run in pinned Docker/Compose environments. Network/service latency is separated from model inference. FAISS in-process latency is not directly compared with remote-service end-to-end latency without decomposition.

23 Discussion

The broader representational question is whether a transformer must interpret a retrieved evidence set through a fabricated one-dimensional global sequence. Local order inside a document is usually meaningful; structural order among sections can also be meaningful; but serialization order among independently retrieved resources may be arbitrary. PRA makes this distinction explicit because logical identity, physical K/V residency, source coordinates, and request-specific composition coordinates can be represented separately.

The immediate result is conservative. Rebinding independently encoded keys to ordinary packed-RAG positions improves numerical fidelity, but does not recover the contextual history of packed encoding. Resource-relative geometries should therefore be evaluated as explicit alternatives, not described as equivalent coordinate changes.

Pre-RoPE storage makes the boundary more precise. It removes position from the stored representation and lets a request apply the host model's exact rotary contract at materialization time. In an unquantized fixture this reproduces the corresponding blocked packed computation. In a production 4-bit model it remains only approximately equivalent because independently shaped projection and attention calls follow different finite-precision paths. More importantly, neither case reconstructs cross-document hidden-state history. Storage format, runtime arithmetic, and causal composition are three distinct concerns.

24 Conclusion

Paper 3.2 evaluates PRA beneath a conventional RAG stack rather than requiring it to replace mature retrieval. Across five natural cohorts, contiguous native K/V reproduced frozen selected-text behavior exactly and made repeated reuse substantially cheaper than text re-prefill. An exact prefix cache remained the better baseline when the serialized prefix itself repeated.

Changing selections reveal both the opportunity and the hard boundary. PRA reused 28.1% of selected K/V where ordinary prefix caching reused none, but independently encoded resources were not compositionally equivalent to a fresh packed prompt. Position rebinding helped; 50% contextual repair helped more; neither restored exact behavior reliably. Partial materialization found one apparently promising 75% point on the first ten questions, but five-seed replication rejected it as a quality-preserving fixed policy. Exercising the canonical Paper 4.5 record/resolver path and placing MiniLM or BGE reranking inside PRA improves selection but does not overturn that result: records are far less order-sensitive and expose non-prefix reuse, yet lose generated-answer F1 on the powered comparison. The practical architecture is therefore clearer than the final solution: external search can freeze strong evidence, PRA can store and reuse exact contiguous native blocks, and future work must make changed compositions contextually faithful without paying full reprefill.

The pre-RoPE causal decomposition narrows that future work. Rebinding independent records to exact packed coordinates closely tracks a packed computation whose documents are causally isolated, but it does not recreate full packing, and 4-bit execution does not preserve exact B/C numerics. Sparse boundary exchange shows that useful interaction may require only a small fraction of all cross-document edges, but the measured quality frontier is non-monotonic. The next composition mechanism should therefore learn or verify which interactions matter, while treating host RoPE compatibility and quantized-runtime parity as explicit, independently tested contracts.

A Artifact Schema

Committed artifact groups include local and service-backed retrieval, five-seed matched realization, partial materialization, independent composition, repair curves, permutation receipts, and changing-selection reuse. Each group retains raw rows or compressed raw rows plus a manifest. Cross-model scale and full positional-policy artifacts use the same schema. The pre-RoPE causal group additionally records the immutable RoPE contract, source and request positions, document spans, condition-specific masks, cross-document edge counts, per-layer K/V errors, first-step logit diagnostics, cold encode time, request-time binding time, and generation-only timings. Seed is the replication unit; confidence intervals and sign-flip tests are computed over seed-cohort means rather than treating questions as independent runs.

B Initial Service Matrix

Backend	Role	Deployment	Status
BM25 local	lexical reference	in process	measured, 50 questions
FAISS + BGE	dense reference	in process	measured, 50 questions
Elasticsearch	lexical reference	remote Docker	measured, 50 questions
Qdrant + BGE	vector reference	remote Docker	measured, 50 questions
RRF service hybrid	lexical + vector	remote Docker	measured, 50 questions
Weaviate or Milvus	vector/hybrid	remote Docker	deferred
pgvector	DB-integrated vector	remote Docker	optional/planned
Derby	user-requested backend	TBD	identity to resolve